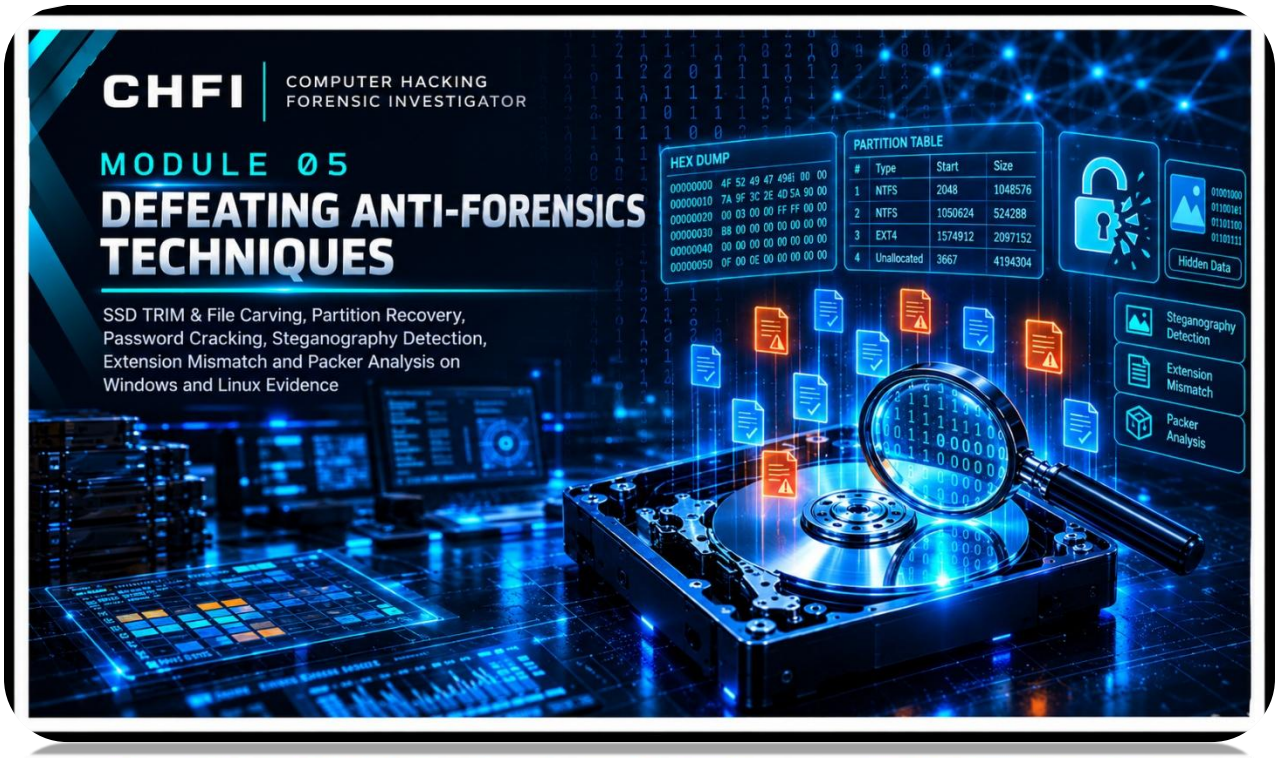


Abdul Wahid

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Defeating Anti-forensics Techniques

Executive Summary

Background. This module documents eight practical labs that identify and defeat anti-forensic techniques used to hinder investigation: SSD TRIM-based anti-carving on Windows and Linux file systems, partition deletion-and-merge concealment, password-protected evidence files, steganographic data hiding in images and audio, file-extension masquerading, and executable packing. Where Module 3 analysed evidence that was handed over pre-processed, this module focuses on evidence deliberately altered or obscured by a suspect, and the techniques used to recover the ground truth underneath. The work was carried out in a virtual training lab, and no real case data was involved.

Environment. The lab used a Windows Server 2025 VM hosting Autopsy, R-Studio, pwdump7, Passware Kit Forensic, Advanced Archive Password Recovery, Advanced PDF Password Recovery, StegSpy, OpenStego, DeepSound, Detect It Easy and Exeinfo PE, alongside the CHFI-Tools evidence share. An optional Ubuntu Forensics VM was used for the zsteg steganalysis step.

Approach. Each lab followed a comparison-and-defeat cycle: establish the anti-forensic condition (TRIM on/off, a hidden partition, an encrypted file, a stego carrier, a renamed file, a packed binary), apply the matching forensic countermeasure, and verify the recovered artifact independently (by preview, hash, size, or content). Recovered or decrypted output was always written to a separate export/recovery folder rather than back onto the evidence.



Principal Findings

1. **SSD TRIM anti-carving (Windows).** Autopsy carving on Windows_Evidence_SSD_TE.dd (TRIM enabled) produced zero recoverable files, confirming TRIM as an effective anti-forensic control. The same carve against Windows_Evidence_SSD_TD.dd (TRIM disabled) recovered carved files (e.g., f0475816.gif) from the $CarvedFiles/Deleted Files set, which were previewed and extracted successfully.

2. **SSD TRIM anti-carving (Linux).** The same TRIM-disabled behaviour was confirmed on a Linux file system: Autopsy recovered deleted files (e.g., f0198840.gif, ~31 KB) from Linux_Evidence_SSD.dd, showing the TRIM effect is OS-independent.

3. **Partition deletion-and-merge concealment.** A 14.22 GB partition was created, populated with test files (including Infected.exe and Confidential.pdf), deleted, and merged into C: by extending the volume. R-Studio's detailed sector scan identified the lost structure as a "Recognized" partition matching the original size and fully recovered its folder structure and files to C:\Recovered Partition.

4. **Credential extraction.** pwdump7 extracted LM and NTLM hashes for all local accounts (Administrator, Guest, and others) directly from the live SAM database in Username:RID:LM-Hash:NTLM-Hash::: format, ready for input to cracking tools such as Hashcat or John the Ripper.

5. **Application password cracking.** Four protected file types were cracked with purpose-built tools: a PDF via Passware Kit Forensic and separately via Advanced PDF Password Recovery (password "test"), a RAR archive via Advanced Archive Password Recovery (password "By2"/"123"), demonstrating that restricting the brute-force character set (digits-only, lowercase-only) materially speeds recovery.

6. **Steganography detection and extraction.** Hidden data was found and extracted across two carrier types and two platforms: StegSpy flagged Model.png (Hideman program detected), OpenStego extracted a concealed Test.txt from Sample.png, zsteg on Ubuntu surfaced multiple hidden text strings in cat.png, and DeepSound extracted Testing1.txt concealed inside Dangerous.wav.

7. **File extension mismatch.** Autopsy's File Type Identification and Extension Mismatch Detector modules flagged 3 files on Windows_Evidence_001.dd whose true MIME signature disagreed with their extension, including tcp-wireshark-file1.trace, which was actually a JPEG image — confirming that signature-level (magic number) analysis is required, since extensions and icons cannot be trusted.

8. **Executable packing.** Detect It Easy and Exeinfo PE both independently identified infected.exe as packed (DiE flagged a high-entropy, compressed ".data" section and overlay; Exeinfo PE flagged the file as not cleanly identifiable and recommended manual unpacking), confirming packing as a viable anti-analysis technique that static signature tools can still flag even without fully unpacking the binary.

Limitations

- TRIM state was taken as given from the supplied image names (TE/TD) rather than independently verified against the live drive controller.
- Passware Kit Forensic, Advanced Archive Password Recovery and Advanced PDF Password Recovery were run in demo/limited brute-force mode with narrowed character ranges chosen for speed, not exhaustive real-world ranges.
- R-Studio's recovery was verified by folder/file presence and size, not by hashing recovered files against the originals.
- pwdump7 output was recorded from screen capture only; hashes were not independently verified by re-running a second extraction tool.
- Steganography tools (StegSpy, OpenStego, zsteg, DeepSound) rely on signature- and heuristic-based detection and can miss custom or non-standard concealment methods.
- Packer identification (DiE, Exeinfo PE) was static only; the binary was not actually unpacked and dynamically analysed.

Integrity

Evidence images were opened read-only wherever the tool supported it (Autopsy ingest, R-Studio scan). All recovered, cracked, or extracted artifacts were written to dedicated export folders (Export, Recovered Partition, Desktop output folders) separate from the source evidence, and originals were left unmodified.

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SSD File Carving on a Windows File System

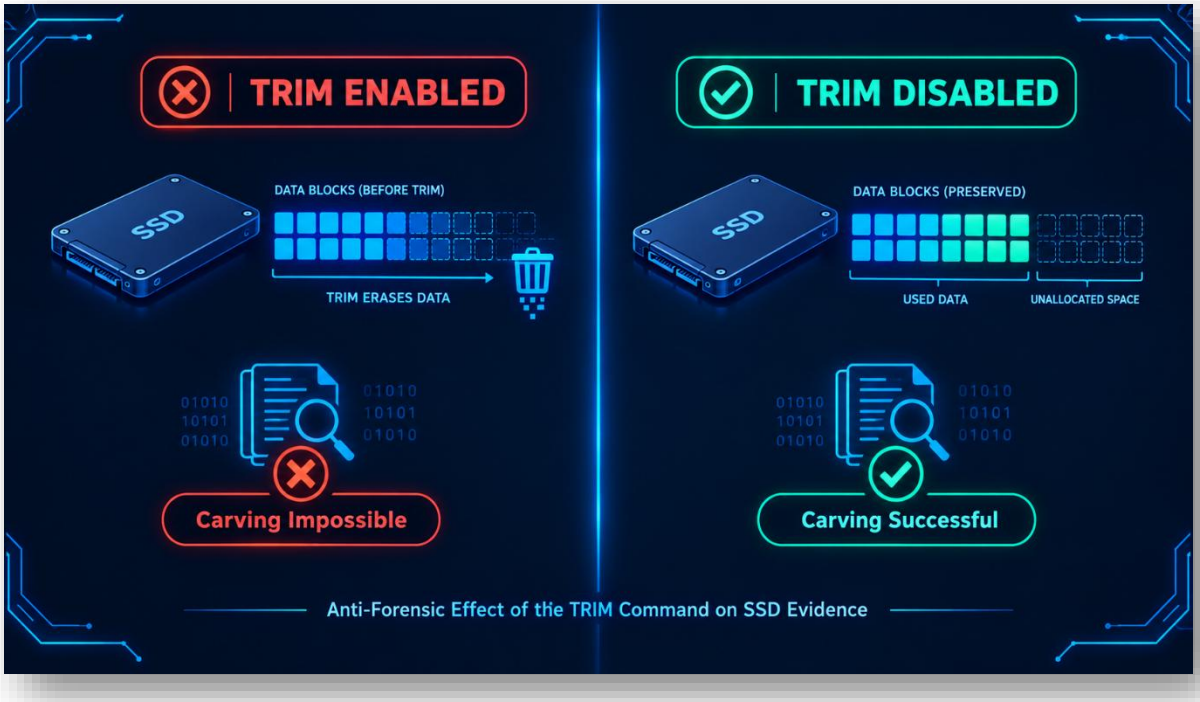
Lab Purpose

The primary purpose of this lab is to investigate the forensic impact of the **TRIM command** on data recovery and file carving within a Windows file system on a Solid-State Drive (SSD).

Specifically, the lab demonstrates how operating system configurations alter evidence persistence:

- **TRIM Enabled:** The storage controller permanently purges deleted data blocks to optimize drive performance, rendering digital forensic file carving impossible.
- **TRIM Disabled:** Deleted files remain intact within unallocated space, allowing forensic investigators to successfully utilize tools like **Autopsy** to carve and reconstruct critical evidence (such as text files and images).

Ultimately, this lab highlights how TRIM acts as an anti-forensic factor and teaches investigators how to identify and validate its state during media analysis.



Lab Setup

Virtual machines needed

- Windows Server 2025 VM

Pre-checks

- Ensure Autopsy and the evidence files (Windows_Evidence_SSD_TE.dd and Windows_Evidence_SSD_TD.dd) are available in CHFI-Tools.
- Autopsy uses significant resources while scanning — avoid using the machine heavily until analysis finishes.

Lab concept: File carving recovers files from unallocated space when file metadata is gone. You use Autopsy on two SSD images of a Windows file system. The first has TRIM enabled (nothing can be carved). The second has TRIM disabled (carved files are recovered).

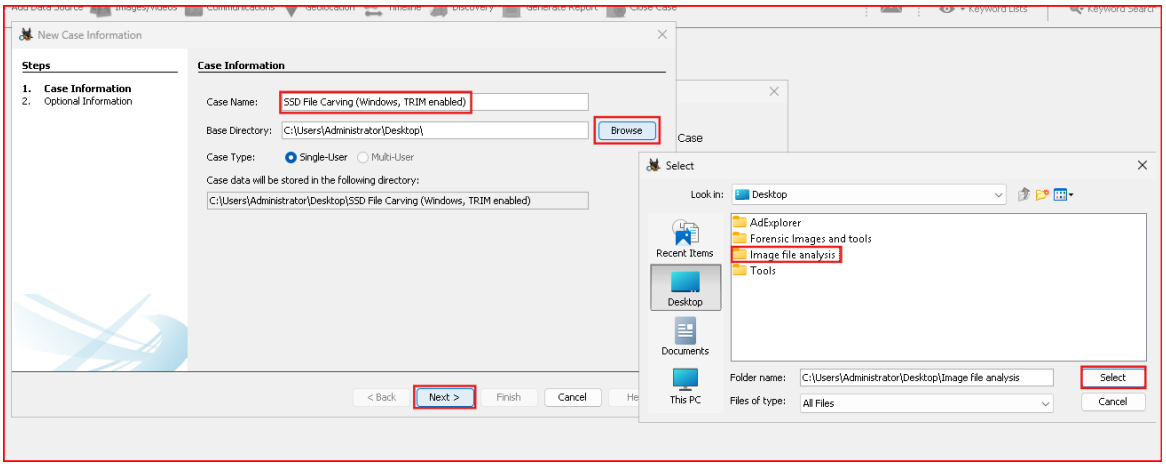
TRIM-enabled image (no carving possible)

Launch Autopsy.

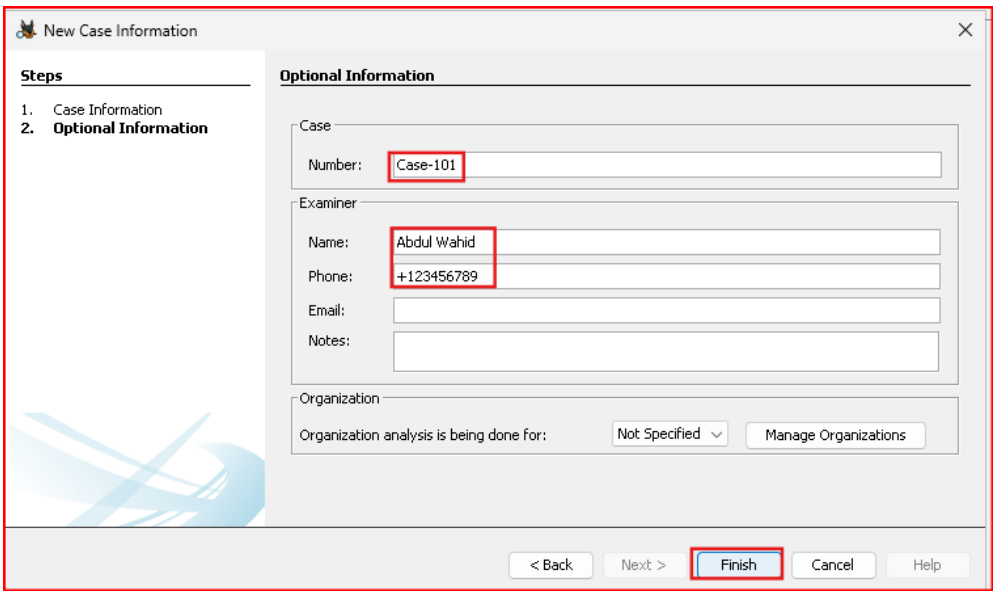
Click **New Case**.

Set the following:

- **Case Name:** SSD File Carving (Windows, TRIM enabled)
- Create a folder on the Desktop named Image File Analysis and select it as the Base Directory.
- Click **Next**.



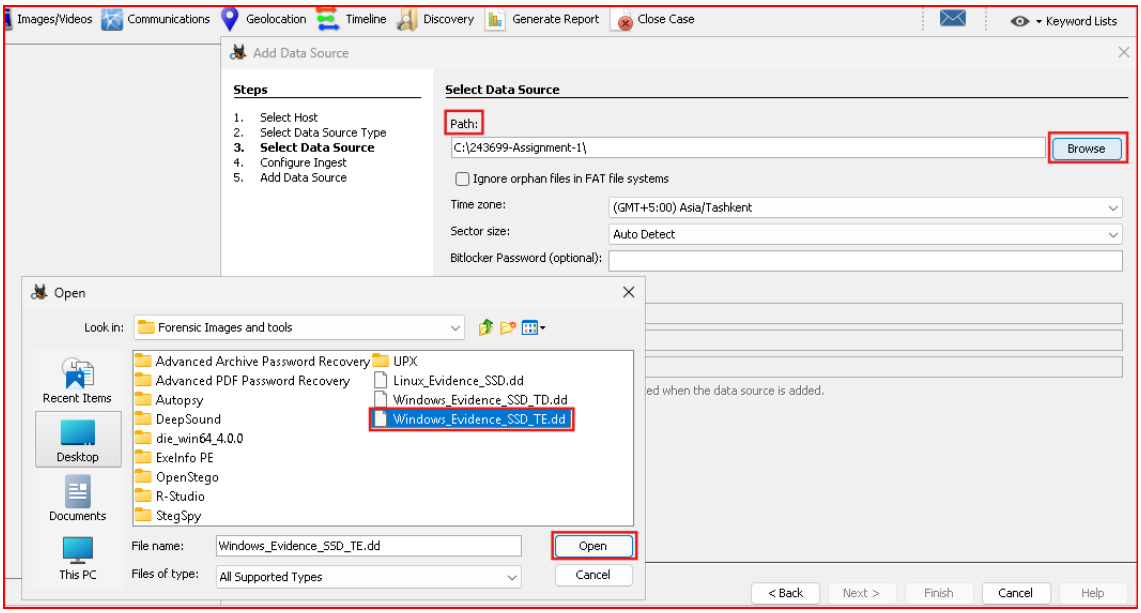
Enter Case Number 101 and your examiner details, then click **Finish**.



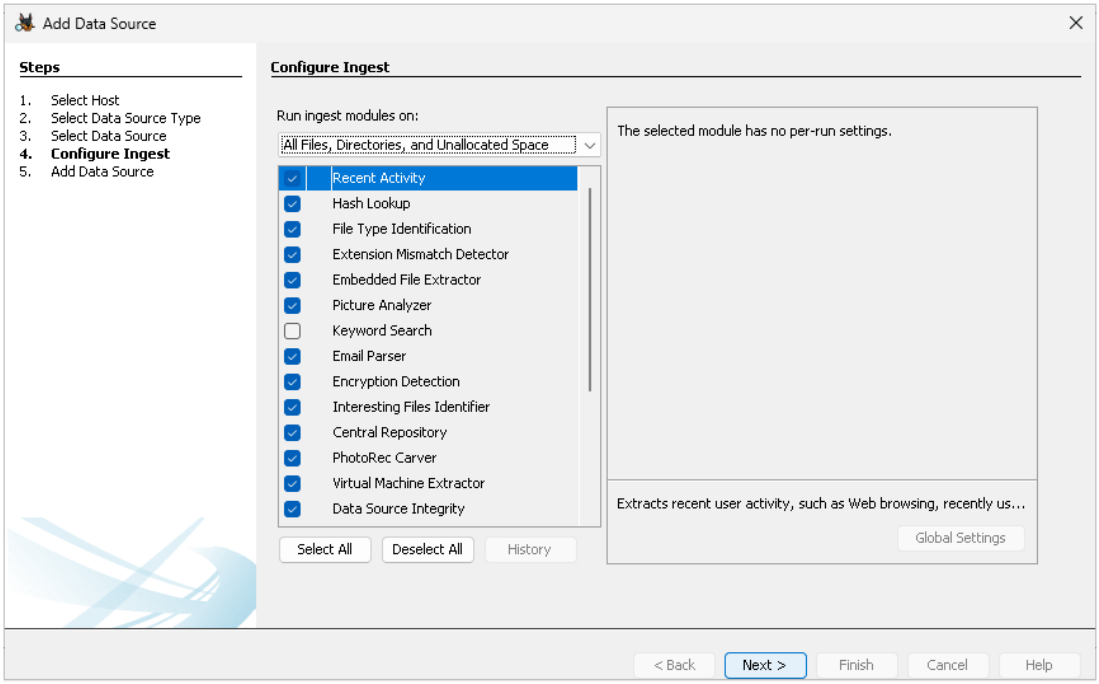
Keep the default host option and click **Next**.

Select **Disk Image or VM File** and click **Next**.

Browse to Windows_Evidence_SSD_TE.dd and add it.

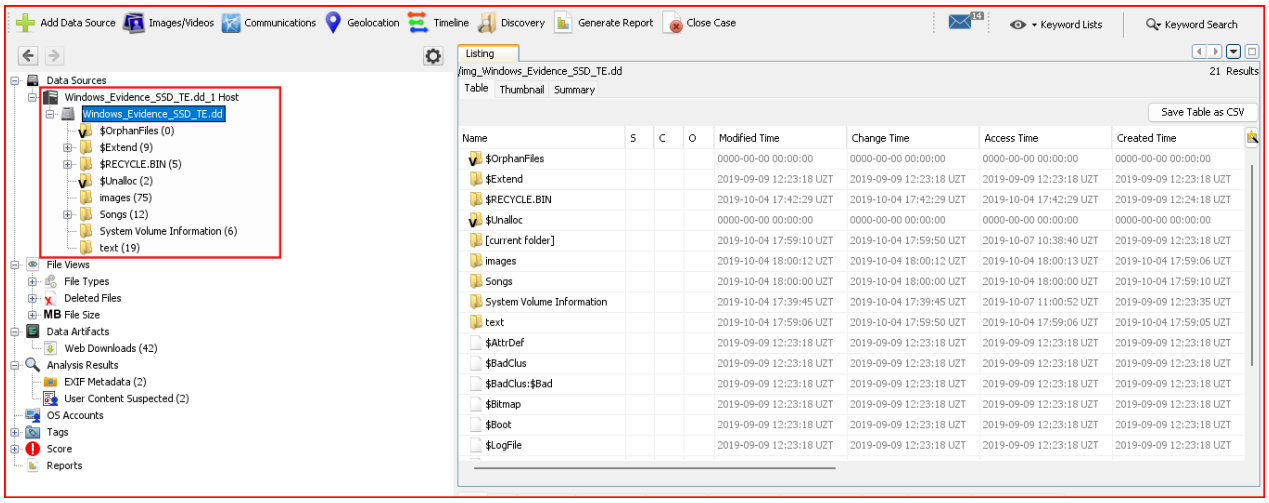


Leave the default ingest modules checked and click **Next**, then **Finish**.



Expand **Data Sources** and select Windows_Evidence_SSD_TE.dd. Wait for the analysis to complete (approximately 10–15 minutes).

Review the results. You will see the normal folder structure, but **no carved files**.



Conclusion: File carving is not possible on a TRIM-enabled SSD image. Close Autopsy.

TRIM-disabled image (carving works)

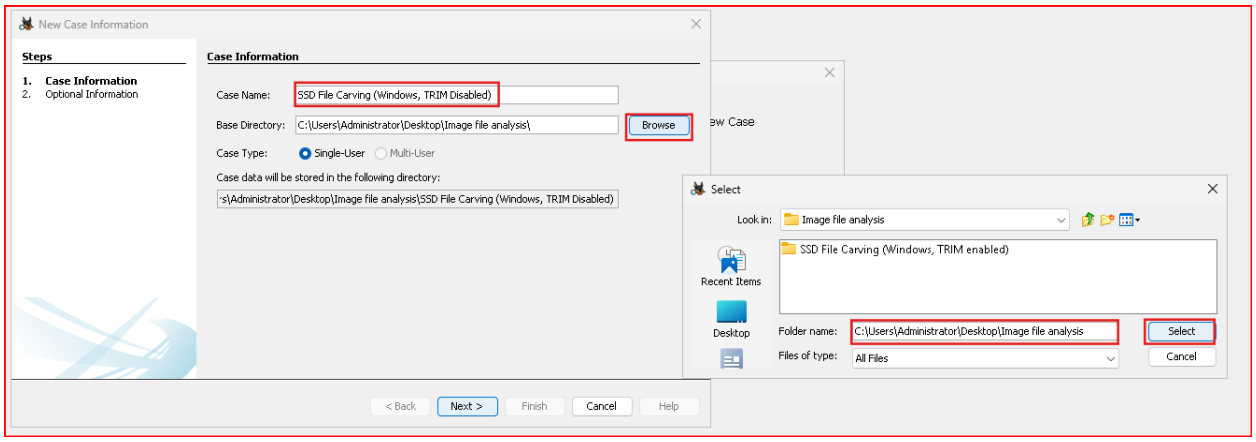
Relaunch Autopsy and click **New Case**.

Set the following:

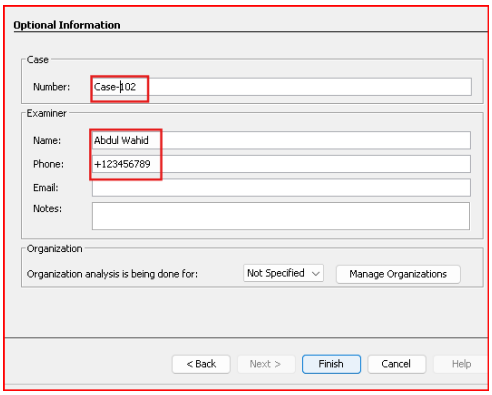
Case Name: SSD File Carving (Windows, TRIM Disabled)

Use the same Image File Analysis folder as Base Directory.

Click **Next**.

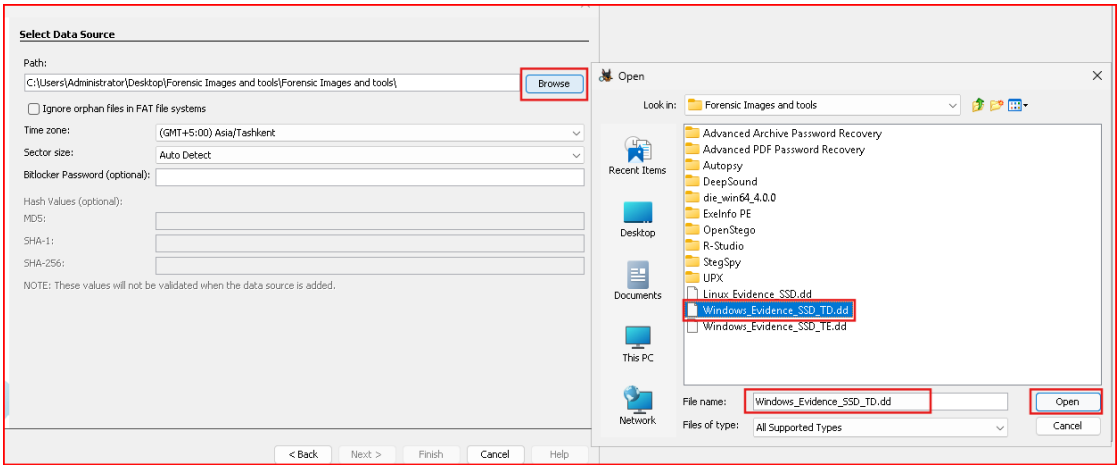


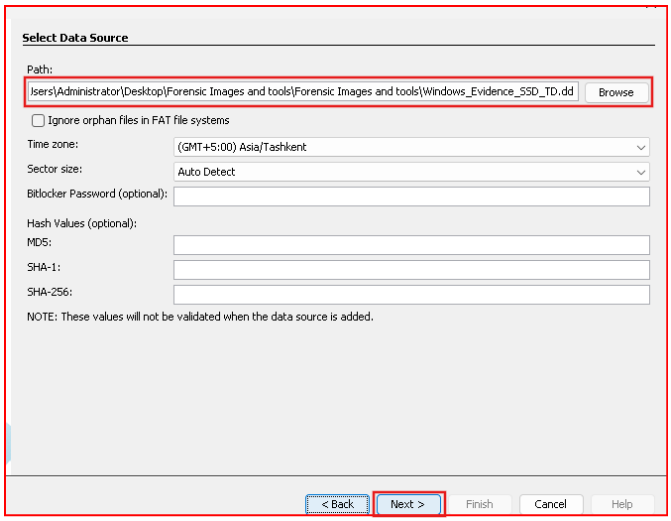
Enter Case Number 102 and your examiner details, then click **Finish**.



Keep the default host and select **Disk Image or VM File**.

Browse Windows_Evidence_SSD_TD.dd and add it.



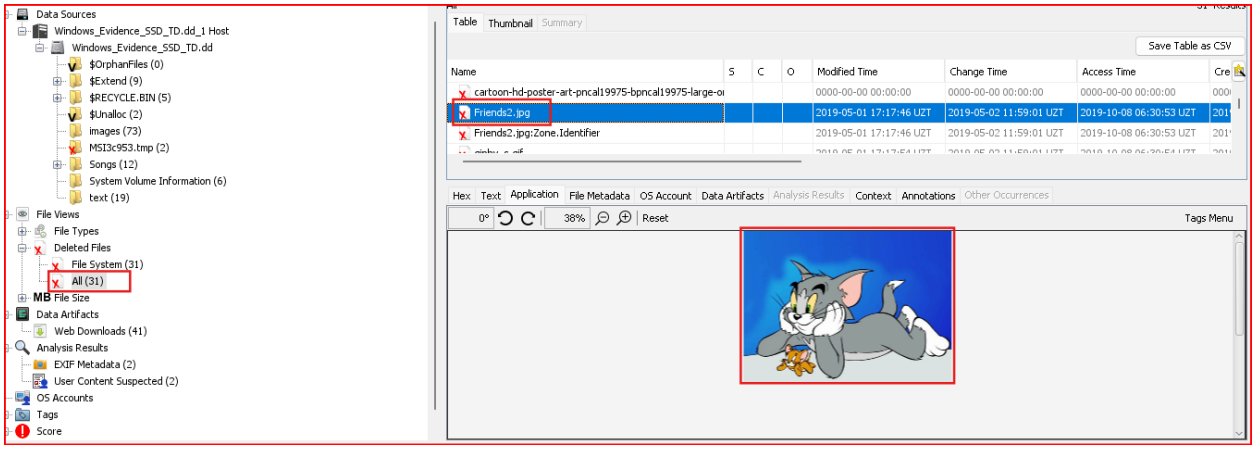


Leave the default ingest modules and complete the addition.

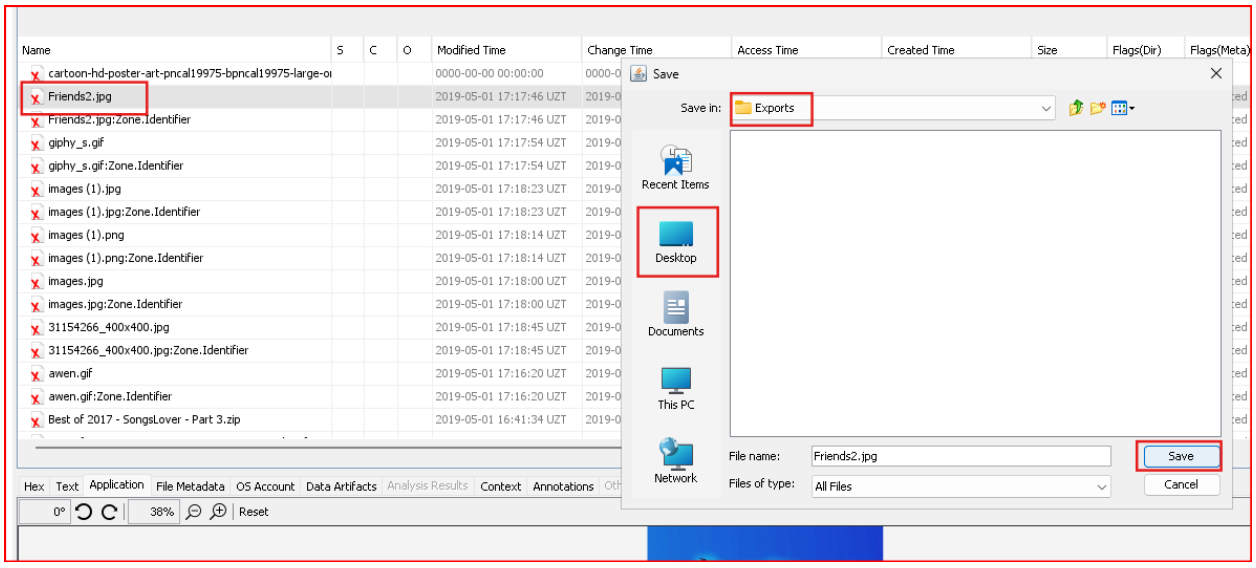
Expand **Data Sources** and select Windows_Evidence_SSD_TD.dd. Wait for analysis (approximately 10–15 minutes).

When complete, open the **$CarvedFiles** folder (or expand **Deleted Files → All**). You should see carved files (e.g. f0475520.txt, f0475816.gif) marked with the cross-and-knife icon.

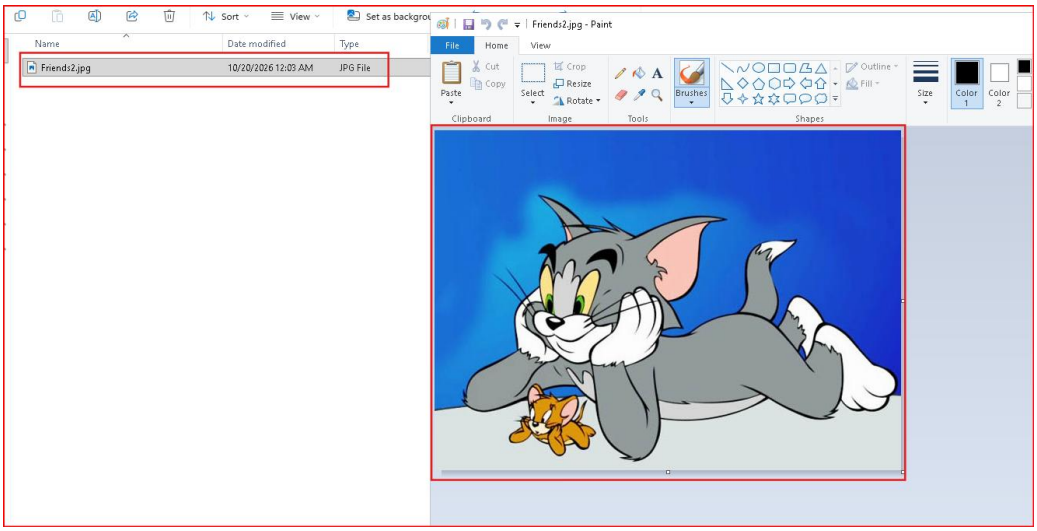
Select a carved file (e.g. f0475816.gif) to preview its content.



Right-click the file → **Extract File(s)** and save it (default location is the case's Export folder).



Navigate to the Export folder and open the recovered file to verify it.



Close all open windows……

SSD File Carving on a Linux File System

Lab Purpose:

The purpose of this lab is to perform digital forensic file carving on a **Linux file system** running on an SSD with **TRIM disabled**. It demonstrates that when TRIM is turned off in a Linux environment, deleted data blocks remain accessible in unallocated space. Using **Autopsy**, investigators can successfully carve, preview, and extract critical evidence (such as images and documents) from Linux media, proving that data persistence principles apply similarly across different operating systems when TRIM is inactive.

Lab Setup

Virtual machines needed

- Windows Server 2025 VM

Pre-checks

- Ensure the evidence file Linux_Evidence_SSD.dd is available in CHFI-Tools.
- The **Image File Analysis** folder on the Desktop (created in Lab 1) will be reused as the base directory.
- Autopsy uses significant resources while scanning — avoid heavy use of the machine until analysis finishes.

Lab concept: Use Autopsy to carve files from the unallocated space of an SSD image of a Linux file system (TRIM disabled), then extract a carved file as evidence.

Launch Autopsy on Windows Server 2025.

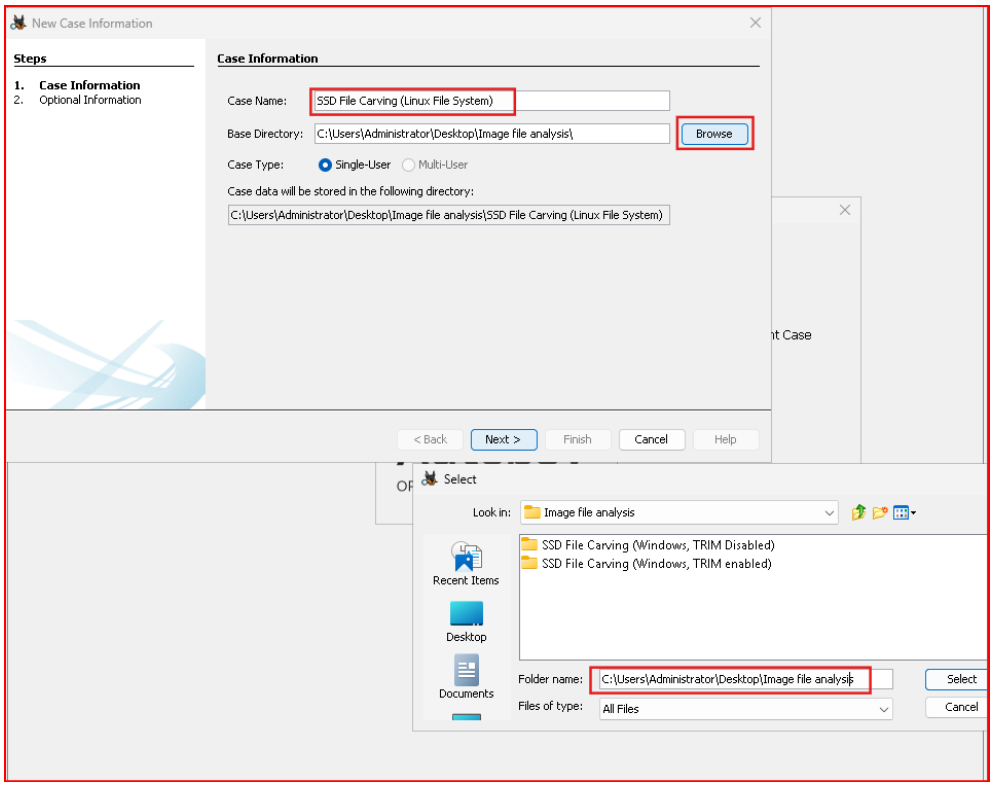
Click **New Case**.

Set the following:

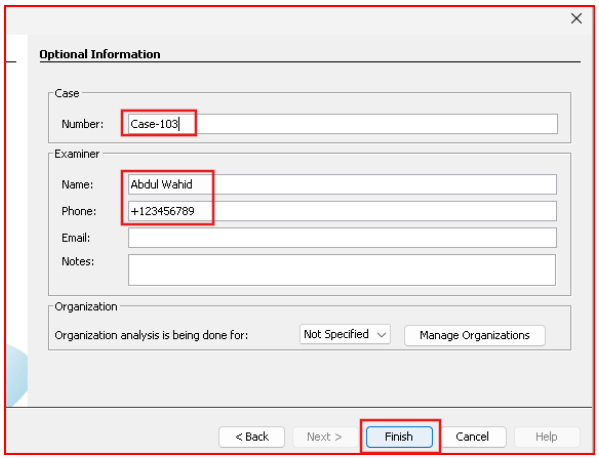
Case Name: SSD File Carving (Linux File System)

Base Directory: the existing Image File Analysis folder on the Desktop

Click **Next**



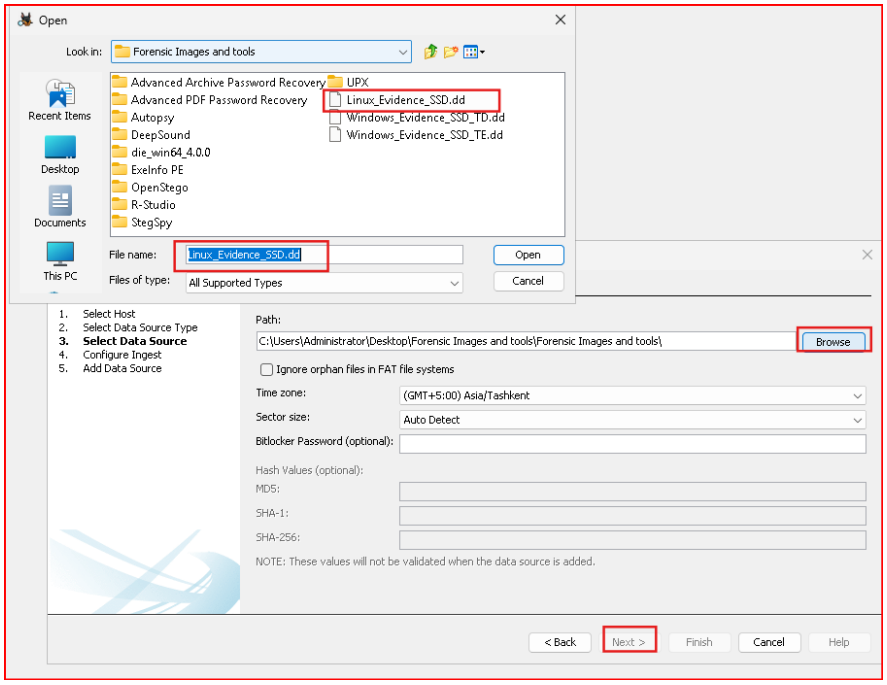
Enter Case Number 103 and your examiner details, then click **Finish**.



Keep the default host option and click **Next**.

Select **Disk Image or VM File** and click **Next**.

Browse Linux_Evidence_SSD.dd and click **Next**.

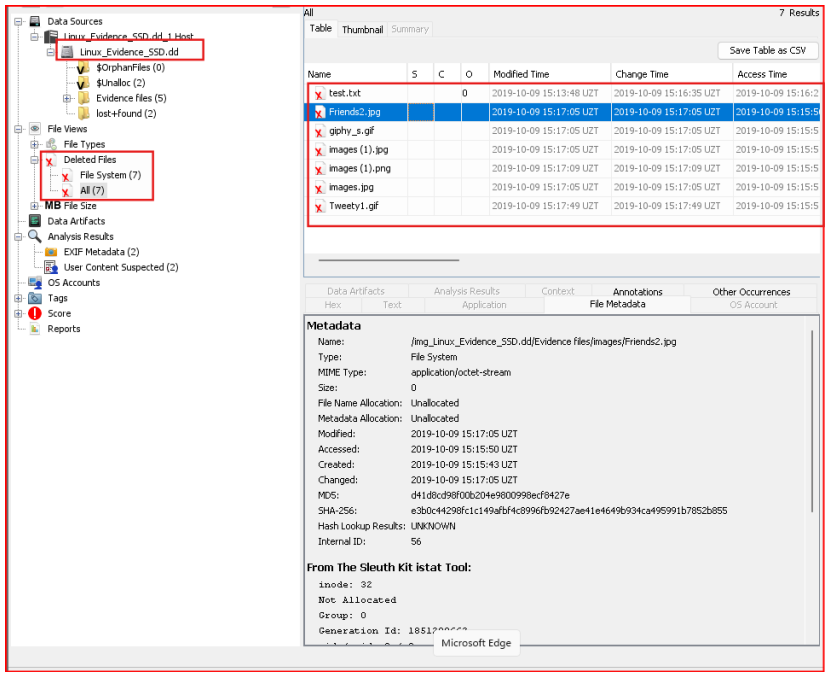


Leave the default ingest modules checked and click **Next**.

When the message "Data source has been added to the local database. Files are being analyzed" appears, click **Finish**.

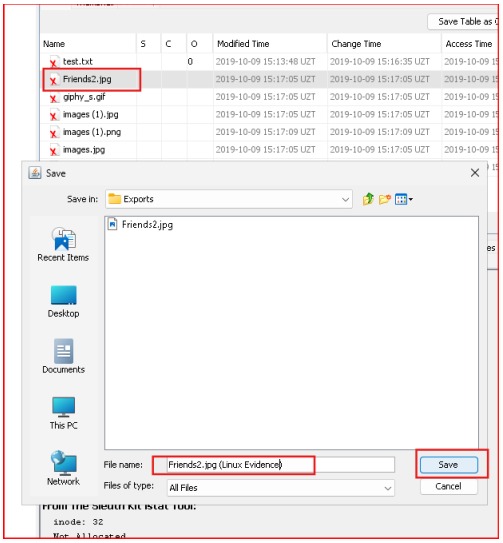
Expand **Data Sources** in the left pane and select Linux_Evidence_SSD.dd. Contents such as Evidence files, lost+found, hello.swp, and test.txt will appear.

Select a Deleted Files (e.g. f0198840.gif). Preview its content in the lower pane (Application, Hex, Text, or File Metadata tabs).

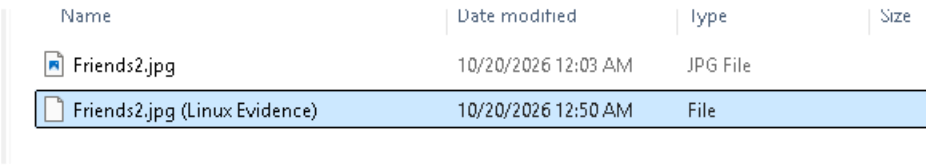


Right-click the file → **Extract File(s)**.

Keep the default Export location and click **Save**. Confirm the pop-up.



Navigate to:
Desktop\Image File Analysis\SSD File Carving (Linux File System)\Export and verify the recovered file (e.g. f0198840.gif, approximately 31 KB).

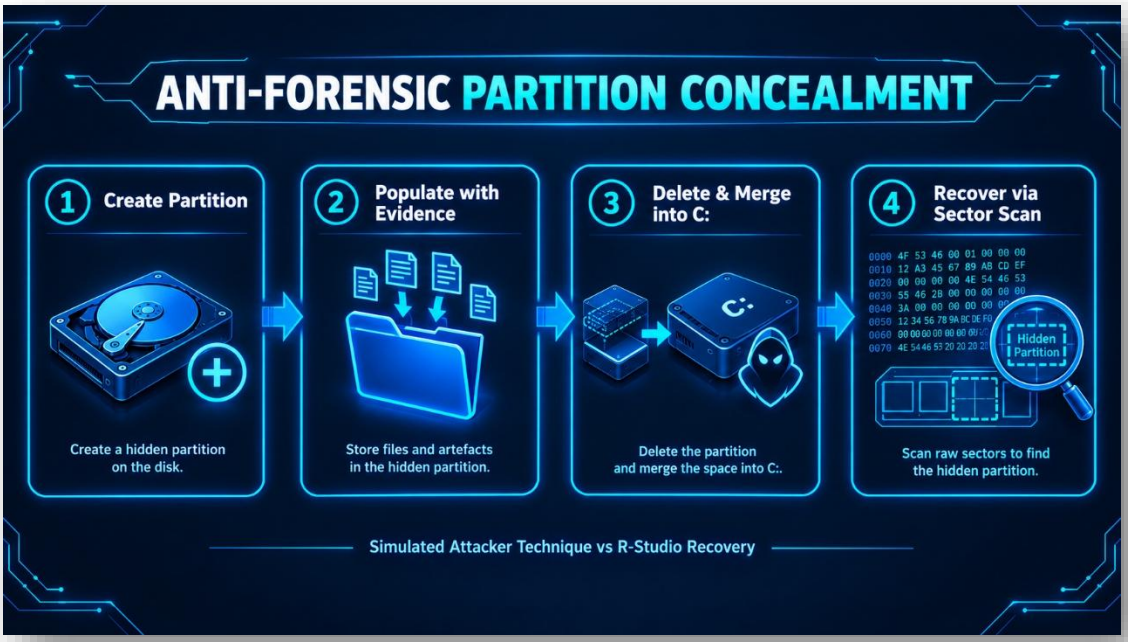


Close all open windows……

Recover Data from a Partition that is Deleted and Merged into another Partition

Lab Purpose:

The purpose of this lab is to demonstrate how to defeat advanced anti-forensic techniques where a suspect attempts to hide data by deleting a partition and merging its unallocated space into an active volume (F: drive). The lab teaches investigators how to perform a raw, detailed sector scan using **R-Studio** to identify lost or overwritten partition structures (Recognized partitions) and successfully reconstruct and extract hidden evidence (such as malware samples and confidential documents) that are invisible to the standard operating system.



Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)

Other requirements

- Administrator's privileges
- At least 15–20 GB free space on the system disk (C:) so you can shrink it and create a second partition
- Internet connection (to download R-Studio Demo if not already present)

Pre-checks

- You are logged in as Administrator.
- Disk Management shows only the normal system partitions + Windows (C:).
- You have **not** written large amounts of data after creating the test partition.

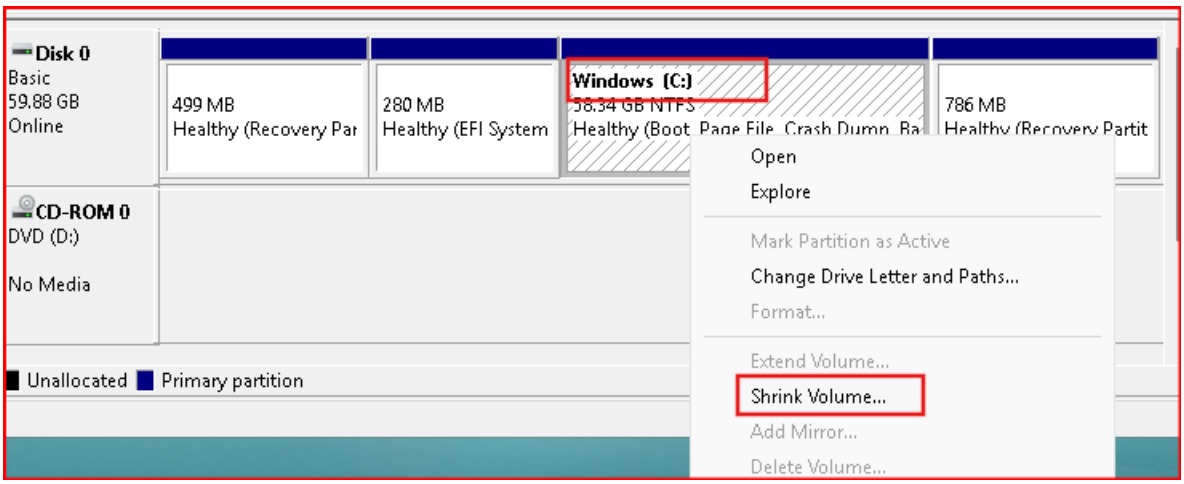
Lab concept: You first create a secondary partition, put test files on it, delete the partition (simulating the attacker), then extend C: over the free space so the deleted partition is "merged". After that you use **R-Studio** to scan the enlarged C: volume, discover the hidden ~15–20 GB recognized partition, and recover its contents.

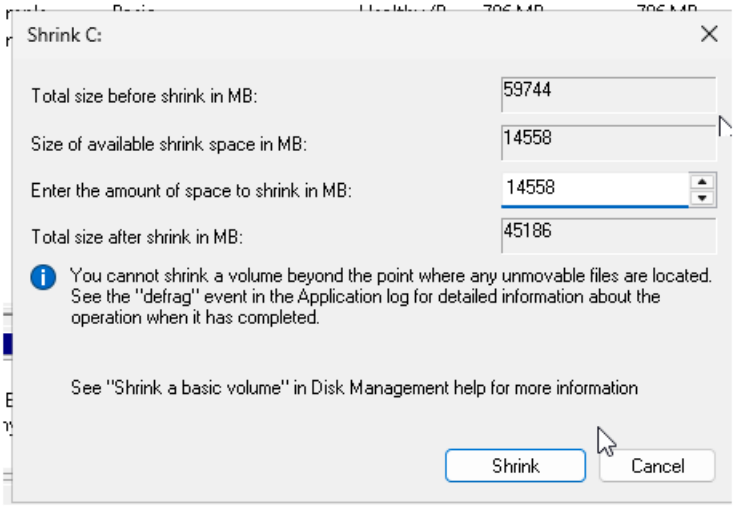
Create the deleted-and-merged partition scenario (simulating the attacker)

Log in to Windows Server 2025 as Administrator.

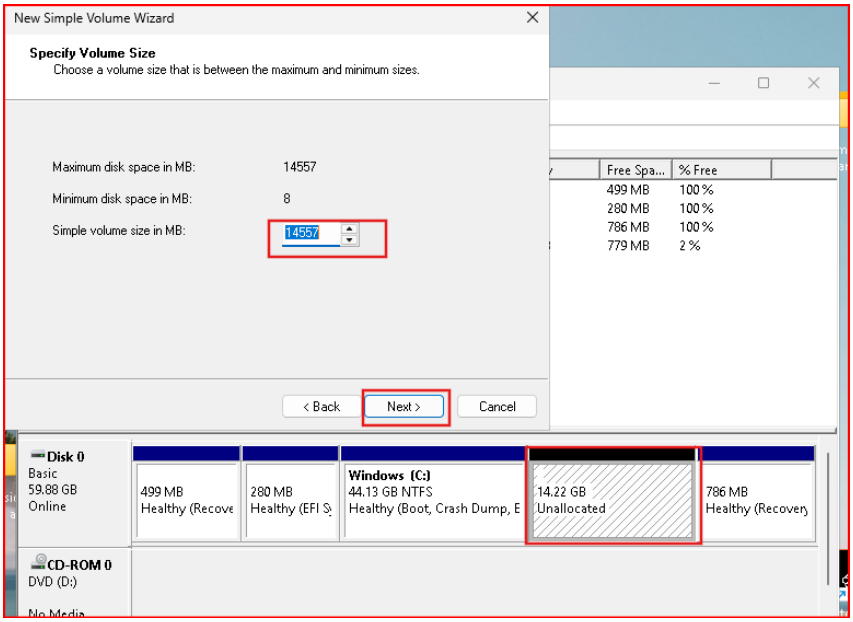
Open **Disk Management** (diskmgmt.msc).

Right-click **Windows (C:)** → **Shrink Volume**.
Enter a value of **14558** and click **Shrink**.

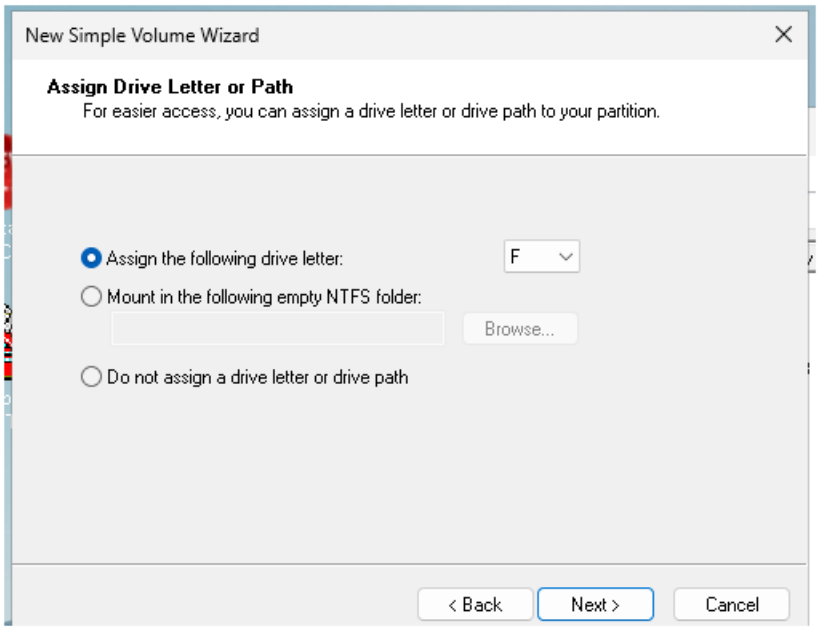




You now see Unallocated space.
Right-click the Unallocated space → **New Simple Volume**.



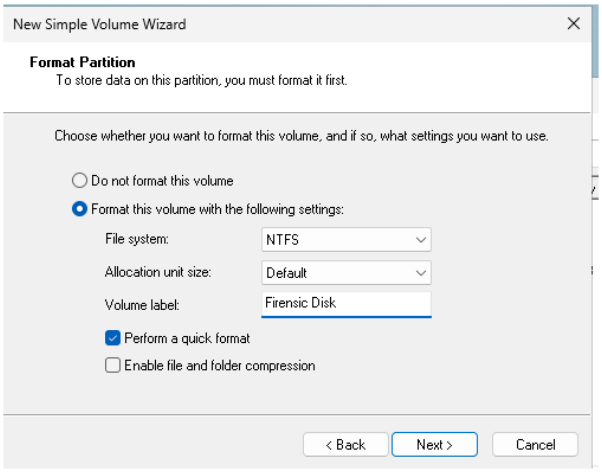
Assign drive letter **F:** (or any free letter).

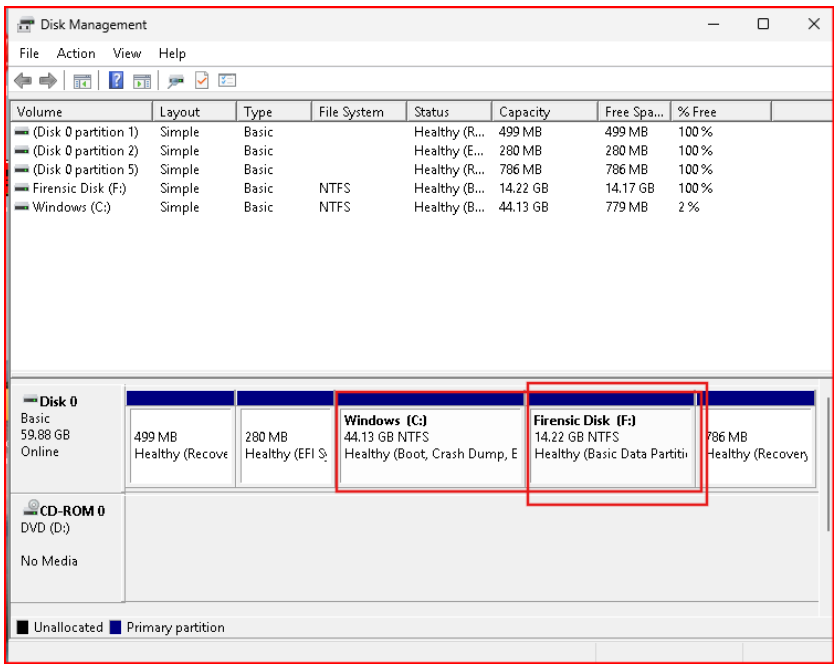


Format as **NTFS**.

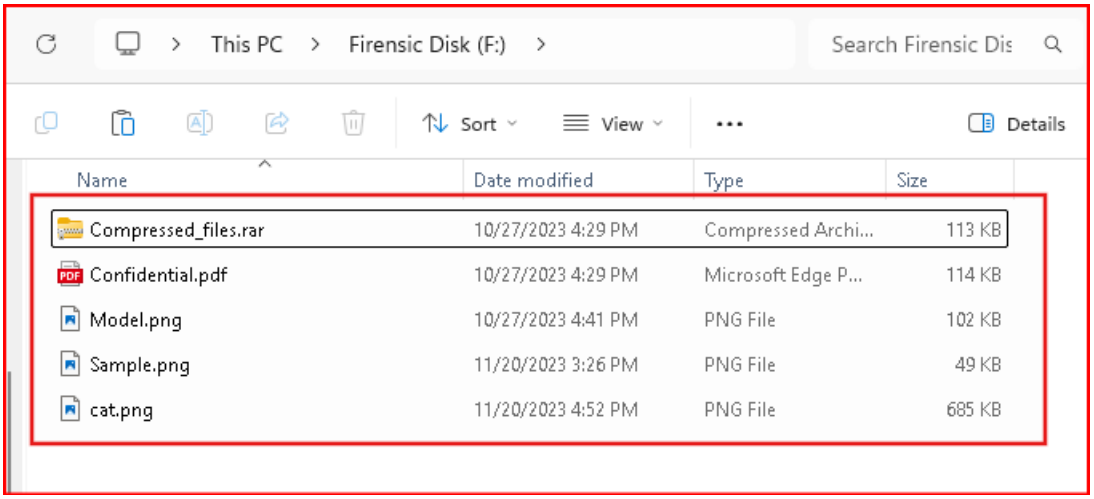
Volume label: **Forensic Disk**.

Finish the wizard.

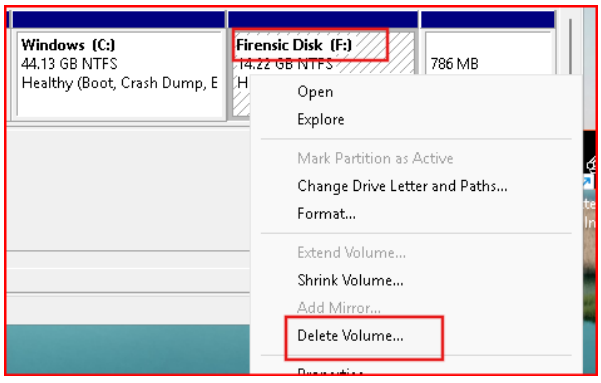




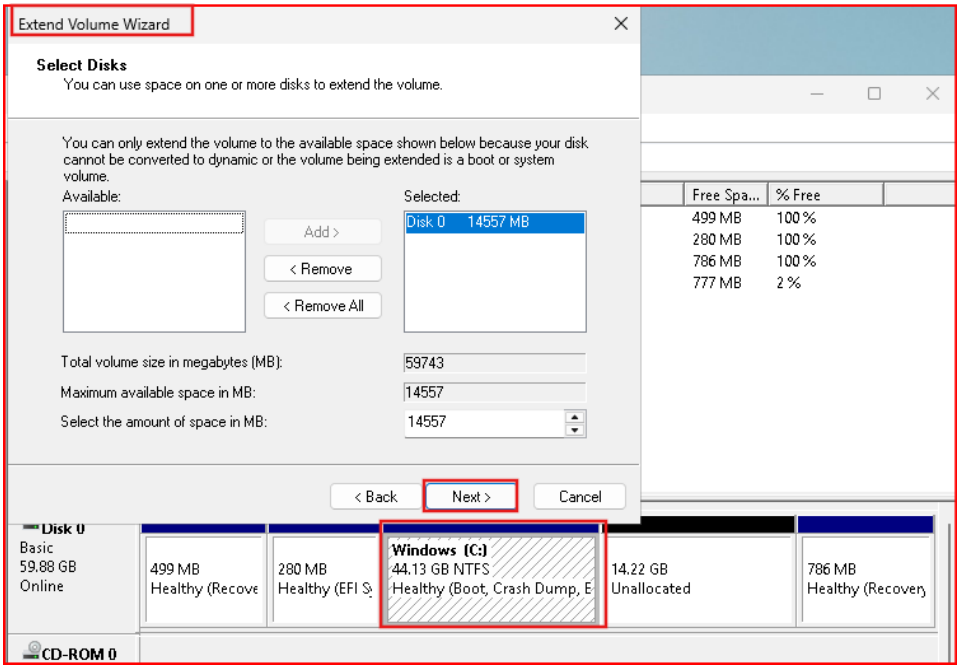
Open the new F: drive and create the following folders:
Audio Files, Image Files, Documents, Text Files.
Copy a few small test files into them (any .mp3, .jpg, .txt, .docx, .pdf, etc.).
Optionally add files named Infected.exe and Confidential.pdf for realism.

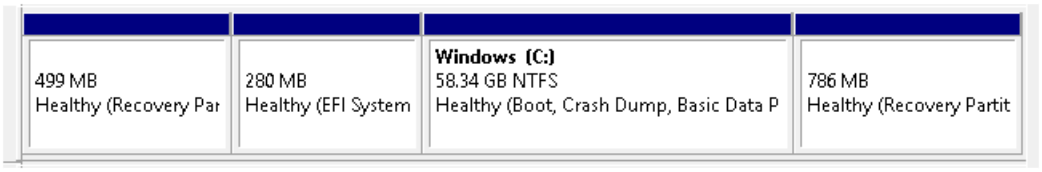


Return to Disk Management.
Right-click the **Forensic Disk (F:)** volume → **Delete Volume** → Yes.
The space becomes Unallocated again.



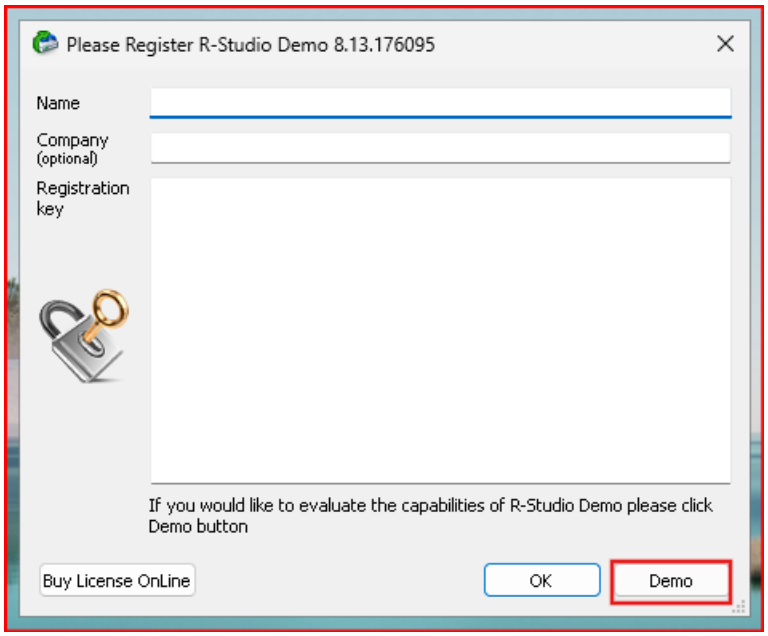
Right-click **Windows (C:) → Extend Volume…**.
Click Next → keep the default selection of the unallocated space → Next → Finish.
Confirm that C: is now larger (original size + the space you just added).
Close Disk Management.





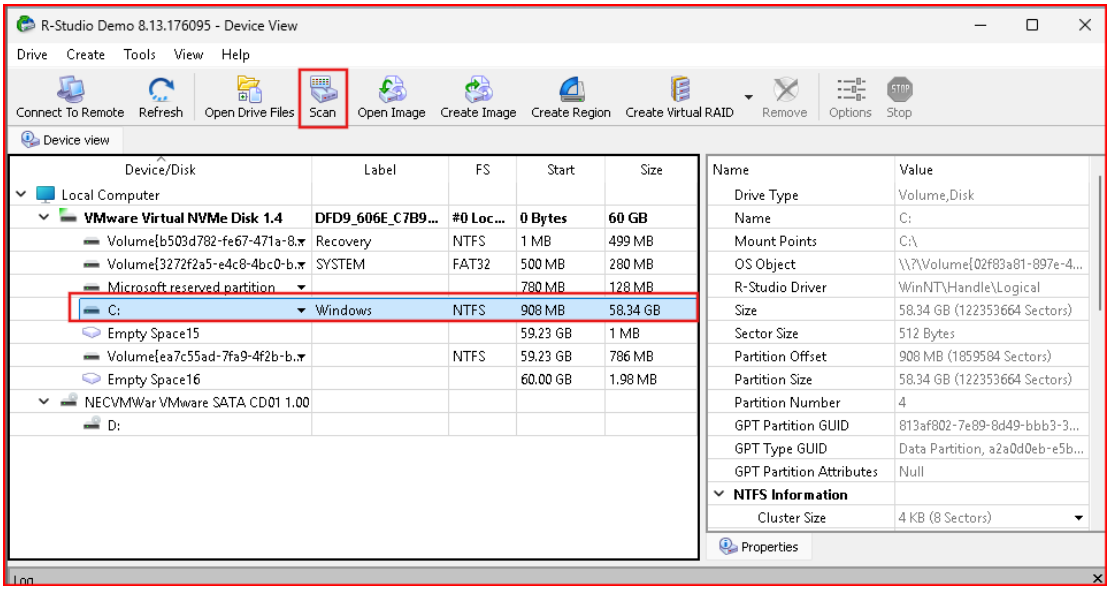
Recover the deleted partition with R-Studio

Download and install **R-Studio Demo** (if not already installed):
Official site → R-Studio → Demo version.
Run the installer, choose English, complete the installation, and launch R-Studio.
When the registration window appears, click **Demo**.



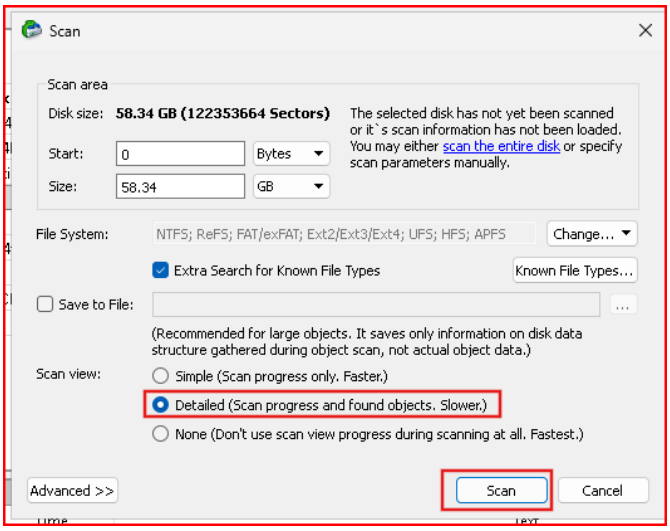
In R-Studio, expand **Local Computer** → select the physical disk that contains C: (or select the C: volume itself).

Click the **Scan** icon on the toolbar.

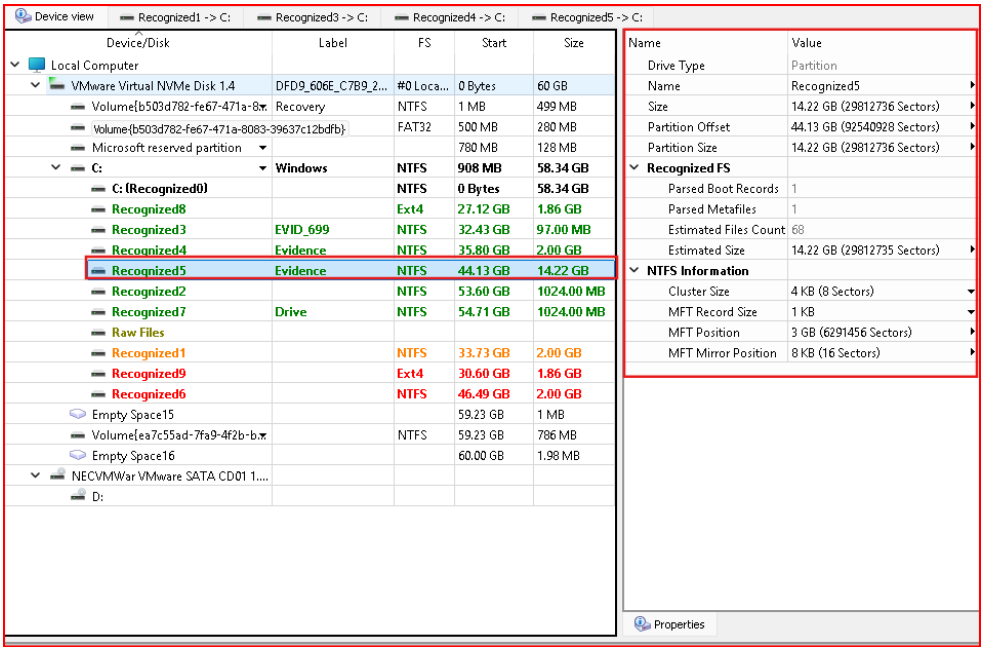


In the Scan window:

- Leave “Scan the entire drive” selected (or adjust if needed).
- Select **Detailed (Scan progress and found objects. Slower.)**.
- Click **Scan**.
  Wait for the scan to finish (progress is shown at the bottom).

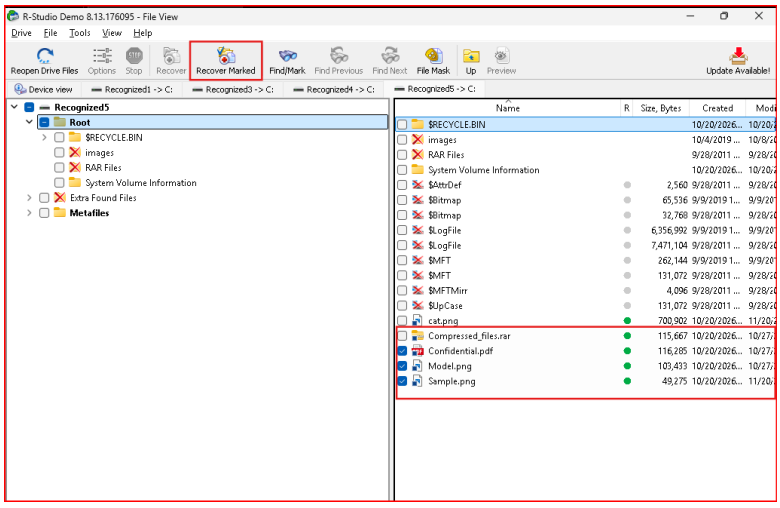


Under the scanned drive you will see one or more **Recognized** partitions.
The one whose size is approximately the same as the partition you deleted earlier (15–20 GB) is the lost partition (often named **Recognized1**).



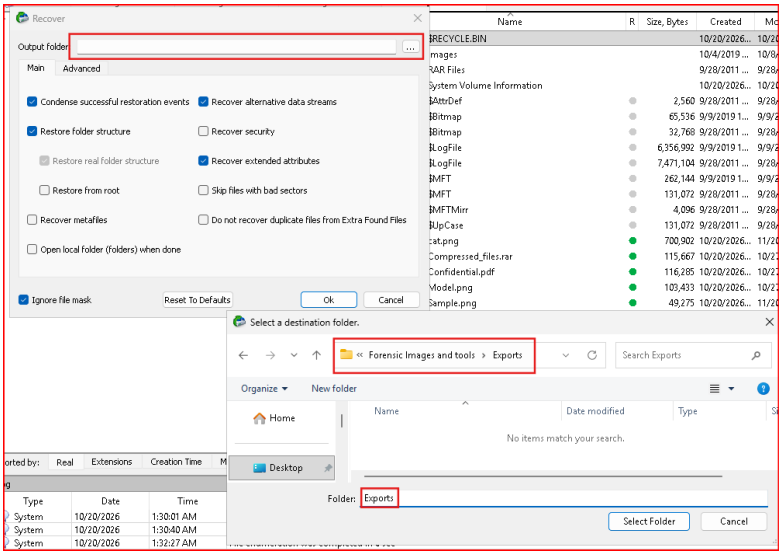
Double-click that recognized partition to open it.
You should see the folders you created (Audio Files, Image Files, Documents, Text Files, etc.) and the test files.

Check the box next to the recognized partition (or next to the folders you want) so everything is selected.
Click **Recover Marked** on the toolbar.

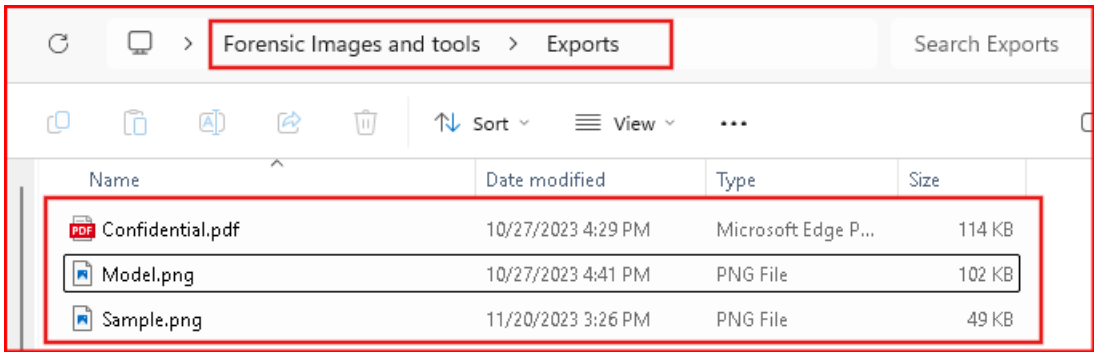


In the Recover window:

- Set **Output folder** to C:\Recovered Partition (create the folder if it does not exist).
- Leave the default options checked (Restore folder structure, etc.).
- Click **OK**.



When recovery finishes, open C:\Recovered Partition\Root (or the folder structure R-Studio created).
Verify that the folders and files from the deleted partition are present.



Close all open window……

Extract Password Hashes from the Target System using pwdump

Lab Purpose:

The purpose of this lab is to extract local user password hashes from the Windows Security Accounts Manager (SAM) database using the **pwdump7** utility. The lab demonstrates how digital forensic investigators can retrieve **LM and NTLM hashes** from a live system with Administrator privileges. Extracting these cryptographic hashes is a crucial step in credential analysis, as they can be documented as evidence or fed into brute-force tools (like Hashcat or John the Ripper) to evaluate password strength or recover actual user credentials during an investigation.

Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)

Other requirements

- Administrator's privileges
- CHFI-Tools available locally (or already copied to Desktop / another drive)

Pre-checks

- You are logged in as Administrator.

Lab concept: You use **pwdump7** to extract the LM and NTLM password hashes from the local Windows Server system. These hashes can later be fed into cracking tools such as Mimikatz, John the Ripper, or Hashcat to recover the actual passwords.

Open Command Prompt as Administrator
Type cmd in the search bar → right-click **Command Prompt → Run as administrator**.
Click **Yes** if the UAC prompt appears.

Navigate to the tool folder

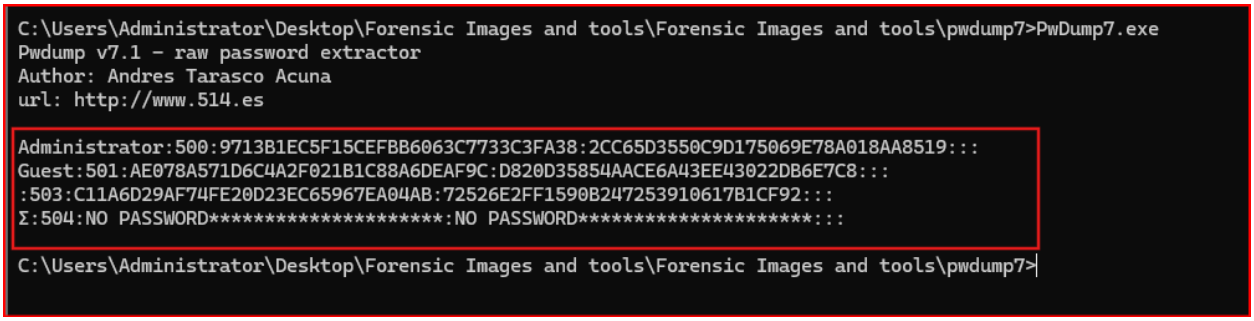
cd C:\Users\Administrator\Desktop\pwdump7

Dump the password hashes

PwDump.exe

(If the executable is named differently, try pwdump7.exe or pwdump.exe)
Press Enter. The command prompt will display the LM and NTLM hashes for each user account on the system.

Note the output
Each line typically shows:
Username:RID:LM-Hash:NTLM-Hash:::
Record these hashes (especially for the Administrator account) for your lab analysis.



Close all open windows…

Crack Application Passwords

Lab Purpose:

The purpose of this lab is to demonstrate digital forensic methodologies for bypassing and cracking application-level encryption on password-protected files. The lab guides investigators through executing local brute-force attacks with specialized forensic software (**Passware Kit Forensic** and **Elcomsoft Advanced Recovery Tools**) by defining custom attack ranges (numeric and lower-case character sets) to optimize recovery speed. Additionally, it highlights the use of automated online decryption platforms (**lostmypass.com**) to recover access to protected spreadsheets, establishing a comprehensive approach to extracting evidence from locked office documents and compressed archives. [1, 2]



Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)

Other requirements

- Administrator privileges
- Internet connection (for the online Excel cracking part)
- CHFI-Tools available locally (or already on Desktop / another drive)

Pre-checks

- You are logged in as Administrator.
- Tools used: Passware Kit Forensic, Advanced Archive Password Recovery, Advanced PDF Password Recovery, and the online tool lostmypass.com.

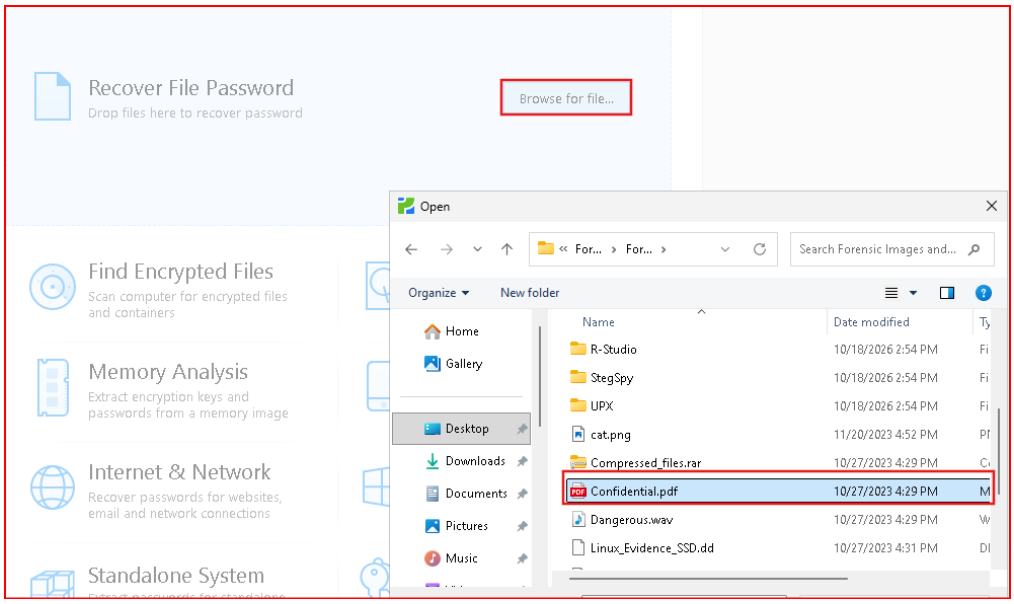
Lab concept: You crack passwords on four types of protected files — a Word document, a RAR archive, a PDF, and an Excel file — using different cracking tools/techniques for each.

Crack a Word document password (Passware Kit Forensic)

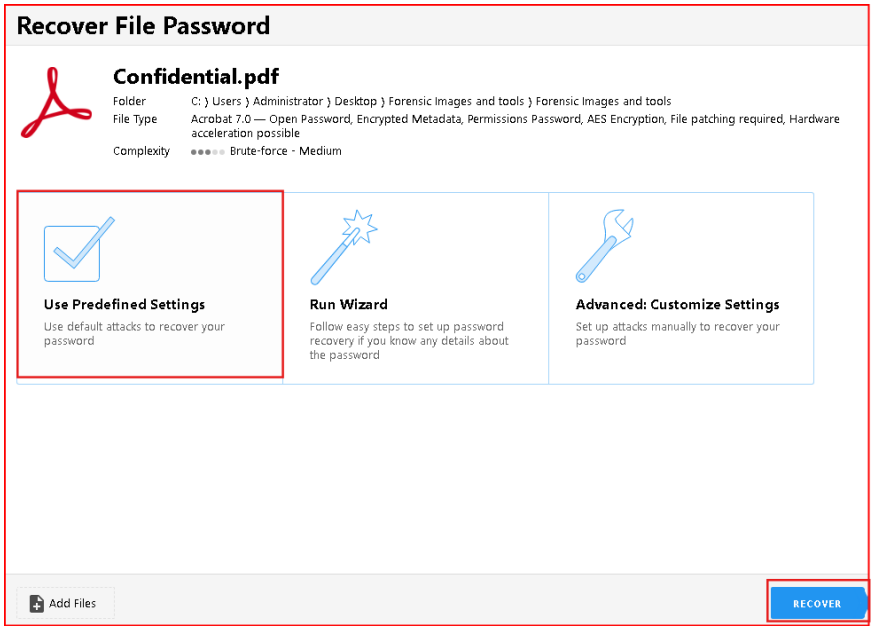
Ensure **Run Passware Kit Forensic Demo** is checked → click **Finish**.

Choose Recover File Password
In the main window, click **Recover File Password**.

Select the target file
Browse to the same Passware Kit Forensic folder, select **Confidential.pdf**, click **Open**.



Start recovery
In the wizard, select **Use Predefined Settings**.

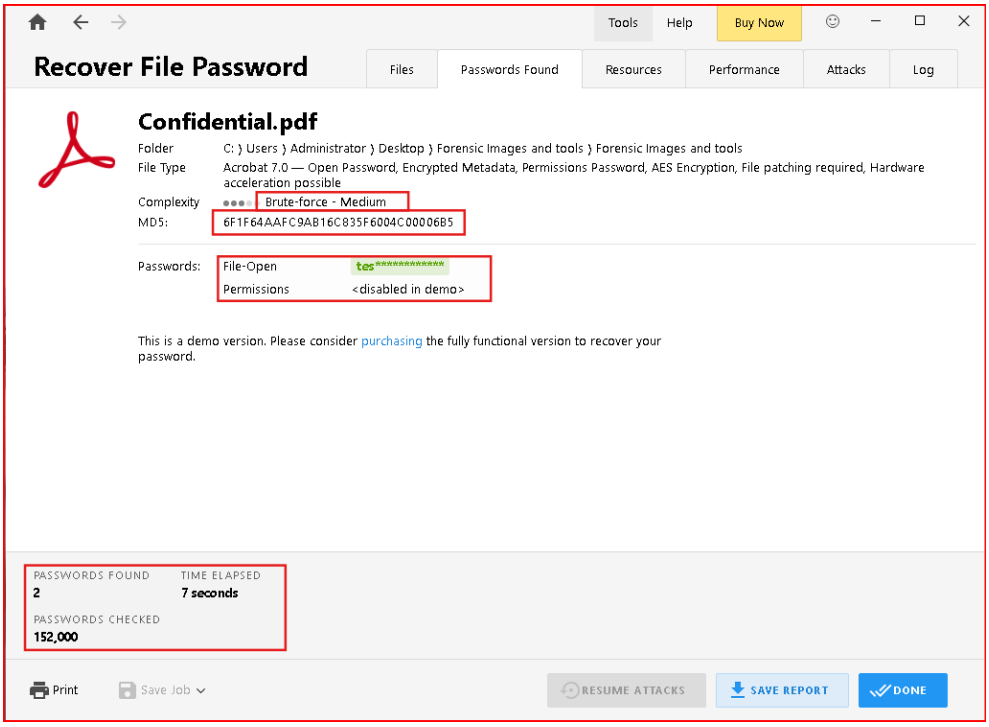


Wait for the crack

The tool brute-forces the password (time depends on complexity).

View the result

The cracked password is displayed (File-Open password is shown; File-Modify usually has none).



Close the application…….

Crack a RAR archive password (Advanced Archive Password Recovery)

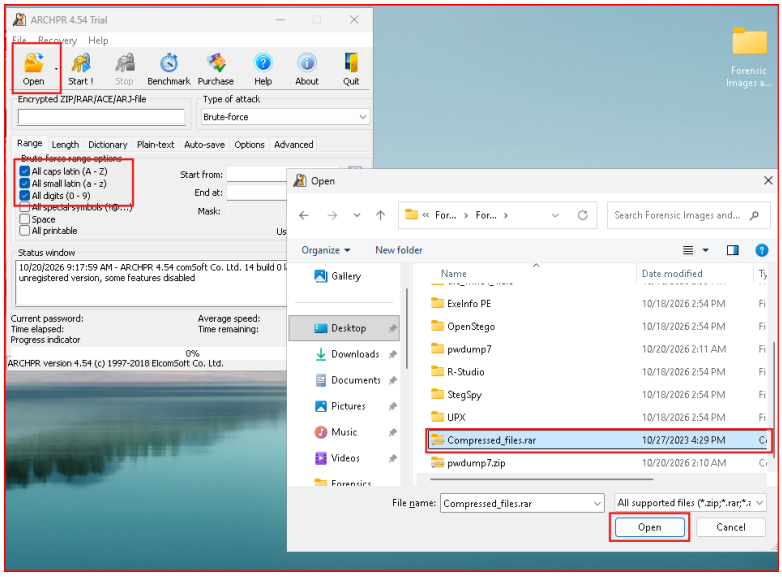
Ensure **Run Advanced Archive Password Recovery** is checked → click **Finish**.

Set attack range

In the **Range** tab, for this demo check **All digits (0-9)** only (leave other options at default).

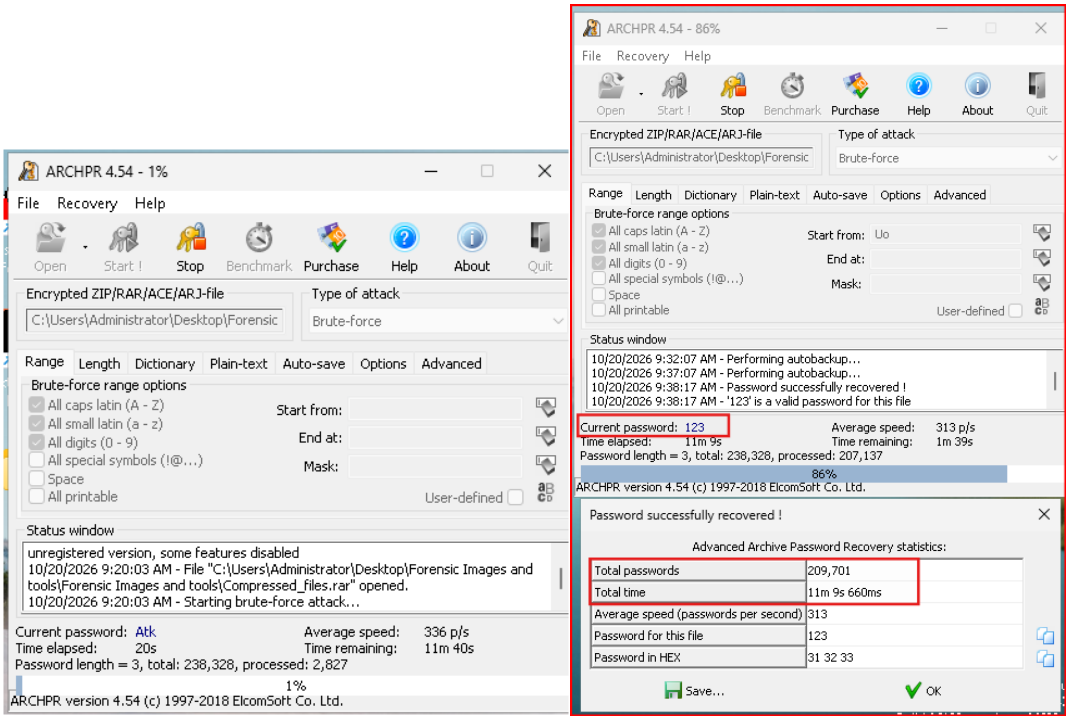
Open the target file

Click **Open**, browse to the same folder, select **Confidential.rar**, click **Open**.



Start the attack

Click **Start!**. The tool brute-forces and displays “Password successfully recovered!” along with the password (commonly shown as **By2**).



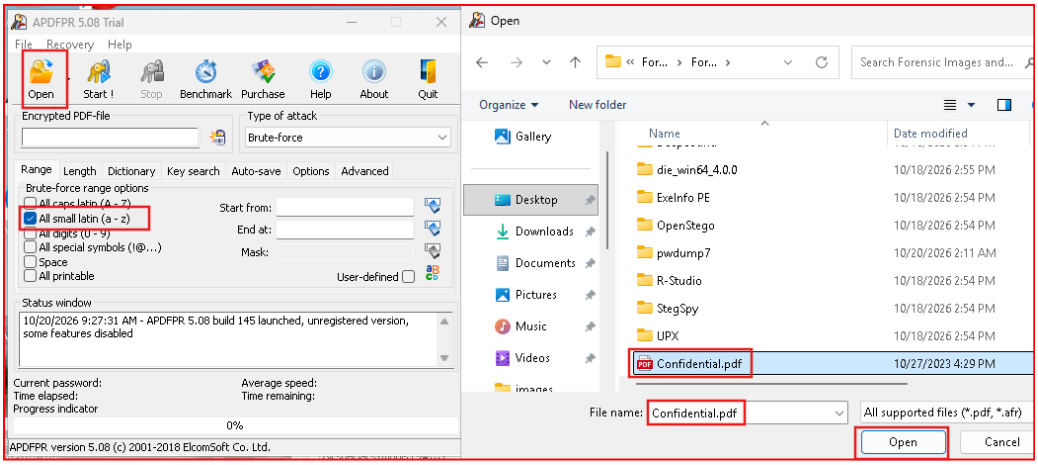
Close the application……

Crack a PDF password (Advanced PDF Password Recovery)

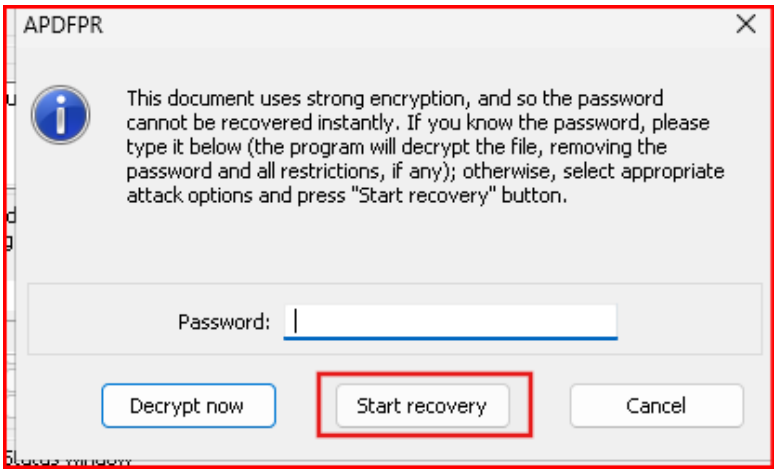
Ensure **Run Advanced PDF Password Recovery** is checked → click **Finish**.
If an Elcomsoft Updater window pops up, close it.

Set attack range
In the **Range** tab, for this demo check **All small latin (a-z)** only (leave others unchecked).

Open the target file
Click **Open**, browse to the same folder, select **Confidential.pdf**, click **Open**.

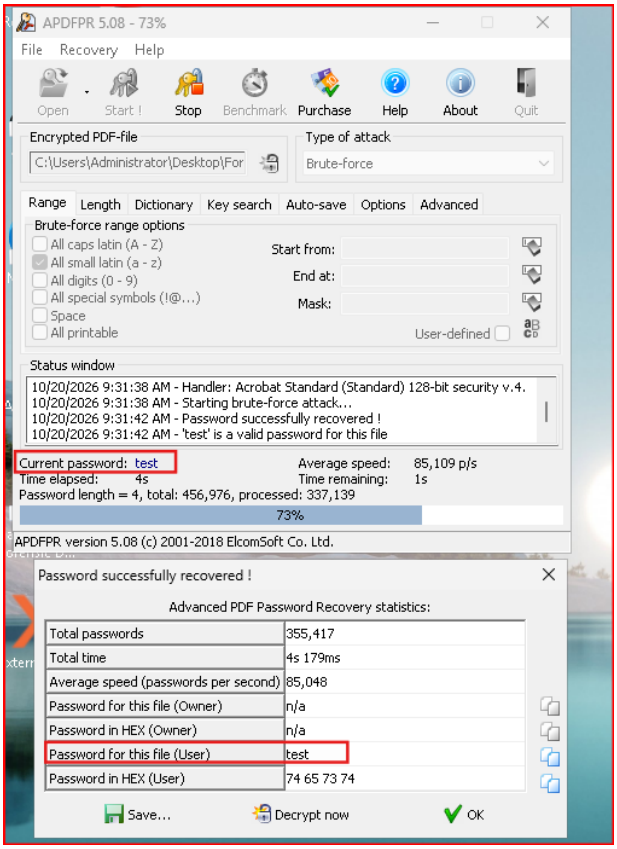


Start recovery
An APDFPR pop-up appears (strong encryption). Click **Start recovery**.



View the result

The tool displays “Password successfully recovered!” with the password (commonly shown as **test**).

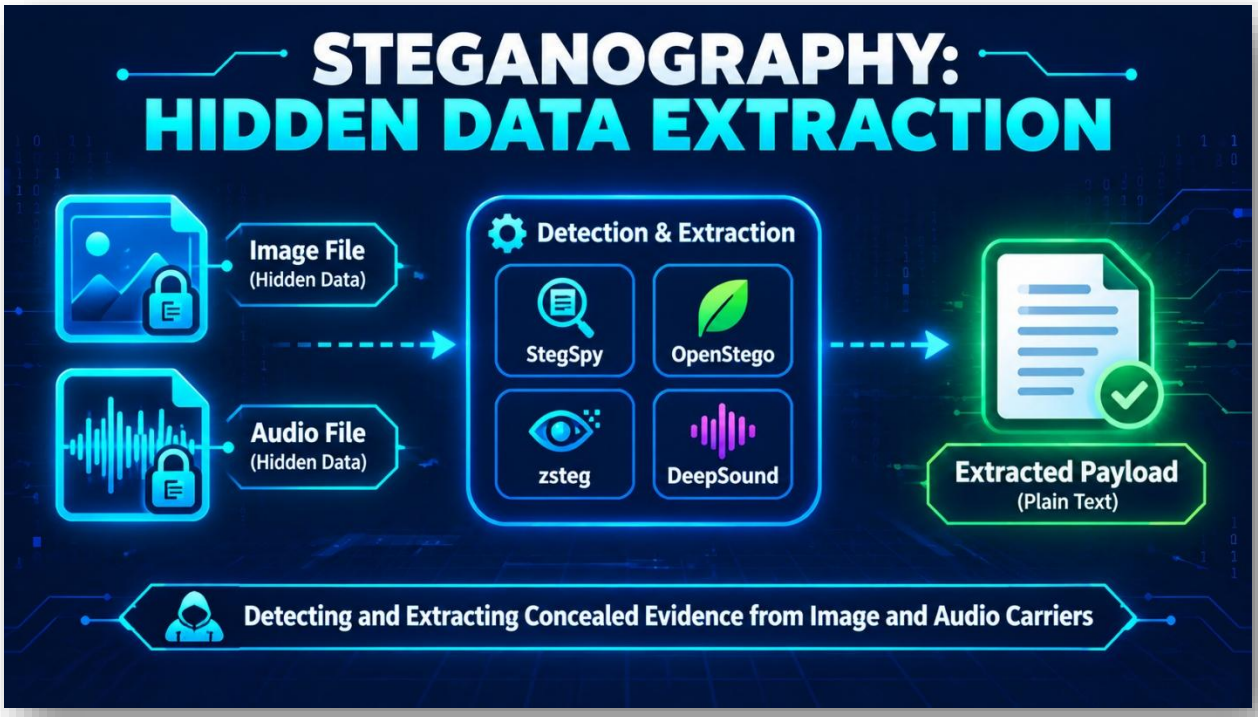


Close the pop-up and the APDFPR tool……

Detect Steganography

Lab Purpose:

The purpose of this lab is to demonstrate advanced digital forensic techniques for detecting and extracting hidden data embedded via steganography inside digital media. The lab guides investigators through analyzing structural anomalies in image formats (.png) and audio containers (.wav) using static analysis utilities like **StegSpy** and bit-level scanning tools like **zsteg**. Furthermore, it provides hands-on experience in recovering hidden evidence payloads (such as concealed text files) using automated extraction platforms like **OpenStego** and **DeepSound**, showing how anti-forensic concealment techniques can be successfully bypassed.



Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)
- (Optional) Ubuntu Forensics VM if you want to run the zsteg part

Other requirements

- Administrator privileges
- CHFI-Tools available locally (or already on Desktop / another drive)
- Internet connection (only if you need to install extra packages on Ubuntu)

Pre-checks

- You are logged in as Administrator.

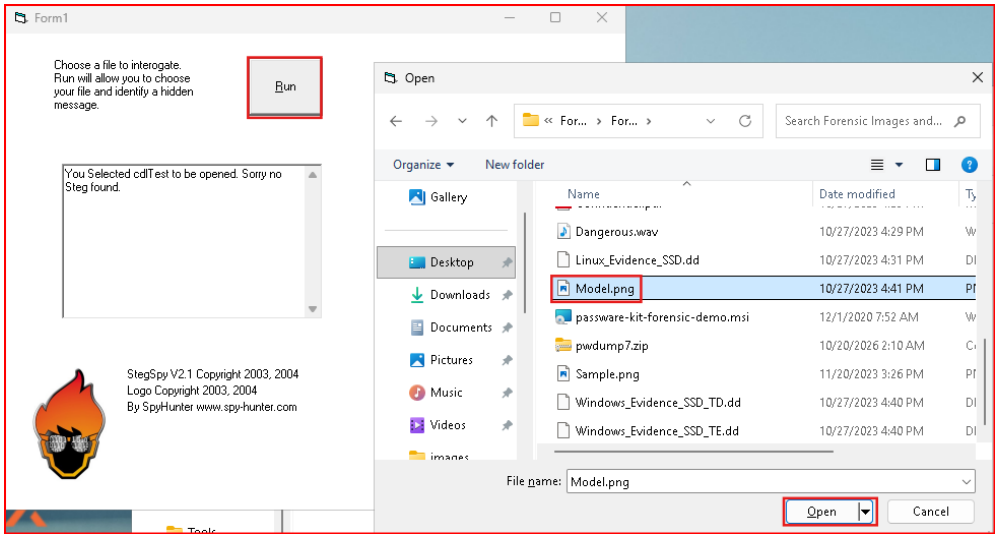
Lab concept: You detect and extract hidden data (steganography) from image and audio files using several tools: **StegSpy**, **OpenStego**, **zsteg** (on Ubuntu), and **DeepSound**.

Detect steganography with StegSpy (Windows)

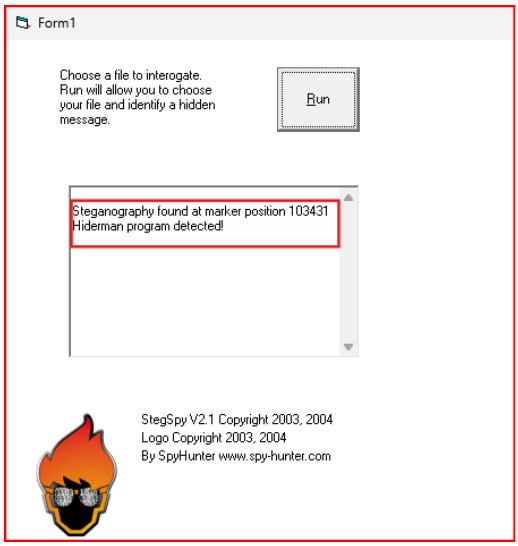
Navigate to:
StegSpy2.1.exe.

Click the **Run** button.

Browse to:
E:\CHFI-Tools\Evidence Files\Image Files
Select **Model.png** → click **Open**.



The tool will detect the steganography technique used (Hideman program detected).



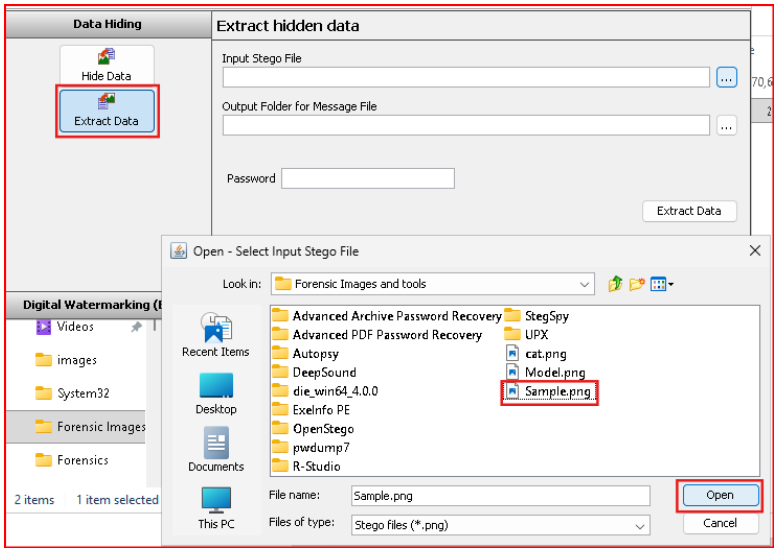
Close StegSpy…….

Extract hidden data with OpenStego (Windows)

Launch **OpenStego** from the Start menu.

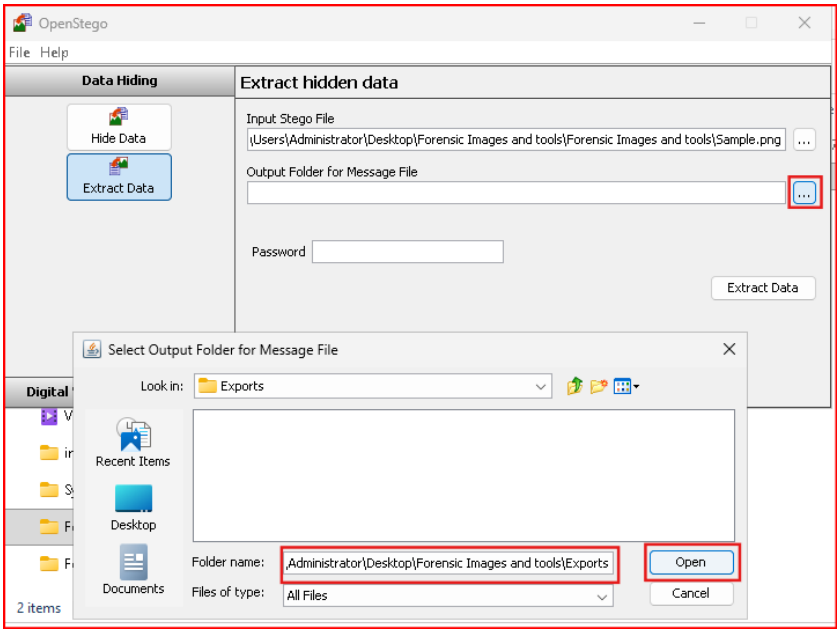
Click **Extract Data** under the Data Hiding section.

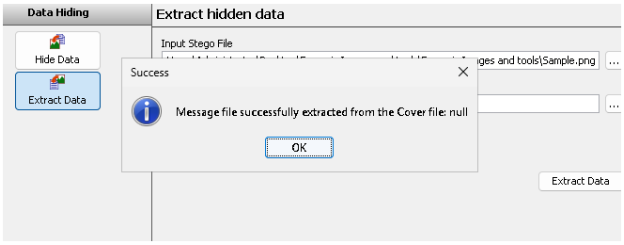
Set: **Input stego file** → E:\CHFI-Tools\Evidence Files\Image Files\Sample.png



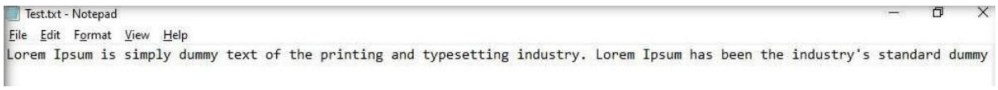
Output folder for message file → Desktop

Click **Extract Data**.
A success message appears and the hidden file **Test.txt** is saved to the Desktop.





Open **Test.txt** to view the extracted content.



Close OpenStego……..

Analyzing with zsteg

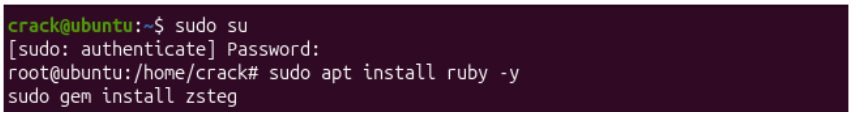
Switch to the **Ubuntu Forensics** VM

Open Terminal and run:

```
sudo su

sudo apt install ruby -y

sudo gem install zsteg
```




```
zsteg cat.png
```



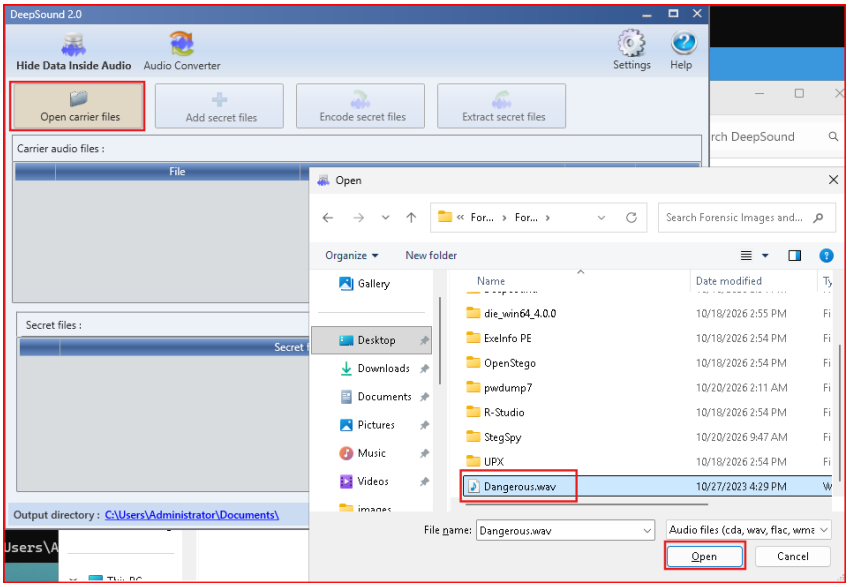
Review the hidden information displayed by zsteg……

Extract hidden data from audio with DeepSound (Windows)

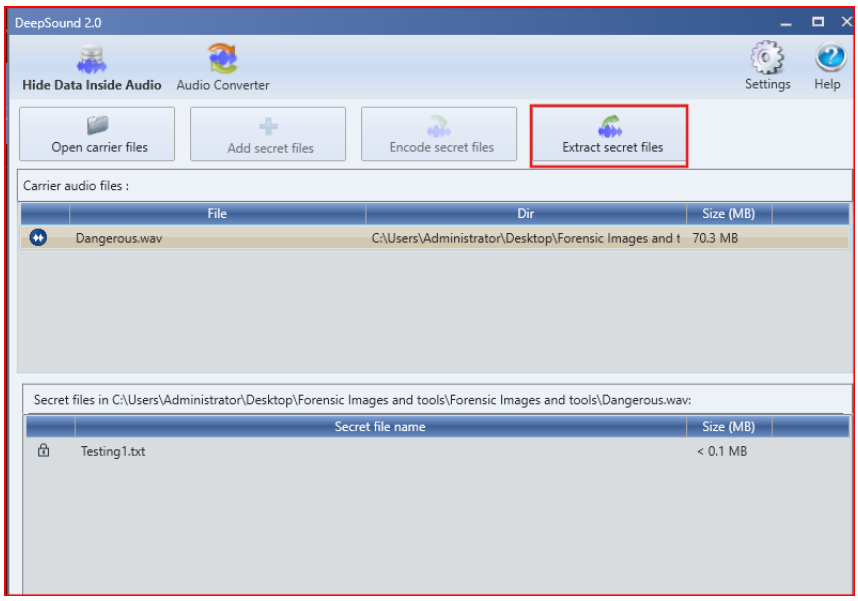
Launch **DeepSound** from the Desktop.

Click **Open carrier files**.

Browse to:
E:\CHFI-Tools\Evidence Files\Audio Files
Select **Dangerous.wav** → Open.

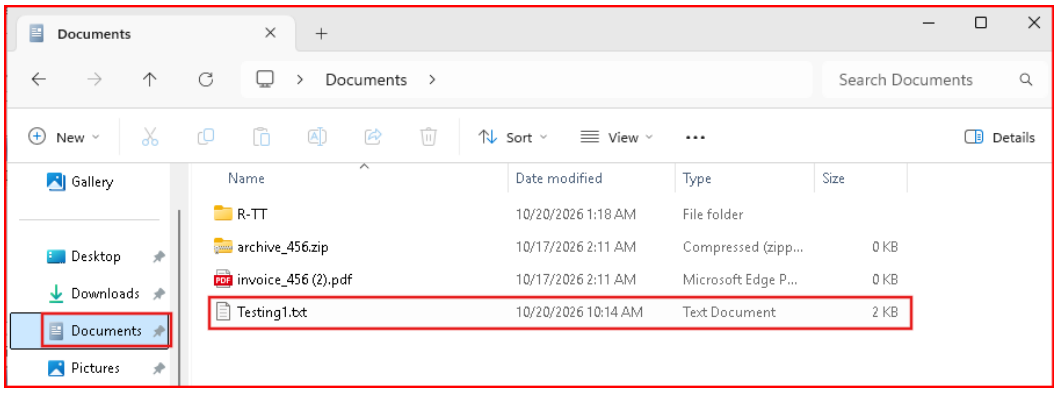


Click **Extract secret files**.



The tool extracts the hidden file (usually **Testing1.txt**) to C:\Users\Administrator\Documents.

Navigate to that folder and open the extracted file to verify the content.



Close all windows……

Detect File Extension Mismatch

Lab Purpose:

The purpose of this lab is to identify and analyze file extension mismatches used as an anti-forensic evasion technique to bypass security controls and mislead investigators. The lab demonstrates how to deploy **Autopsy's File Type Identification** and **Extension Mismatch Detector** ingest modules to analyze the raw binary signatures (magic numbers) of files independently of their superficial extensions. Investigators gain hands-on experience in isolating disguised assets—such as a .jpeg image masked with a .trace extension—proving that file integrity validation at the hexadecimal/signature level is crucial for uncovering hidden evidence during a drive analysis.

Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)

Other requirements

- Administrator privileges
- Autopsy must already be installed

Pre-checks

- You are logged in as Administrator.
- The **Image File Analysis** folder on the Desktop (created in earlier labs) can be reused as the base directory.
- Autopsy uses most of the VM's resources while scanning — avoid using the machine until the scan reaches 100%.

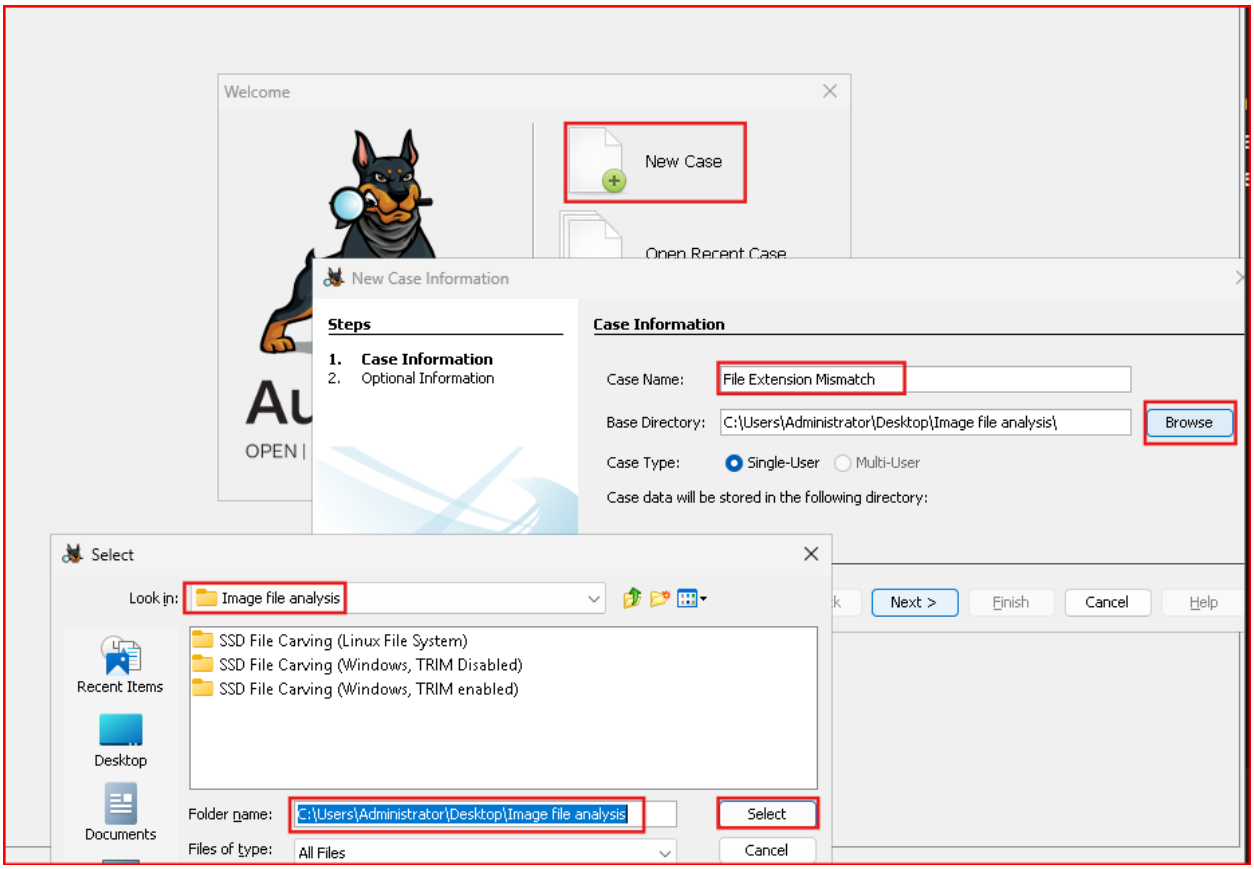
Lab concept: Attackers sometimes rename a malicious file's extension (for example, disguising a JPEG as something else) to evade firewalls, anti-malware tools, and investigators. You use Autopsy's **Extension Mismatch Detector** module to catch files whose real content doesn't match their extension.

Launch **Autopsy**
Double-click the Autopsy shortcut on the Desktop.
In the Welcome window, click **New Case**.

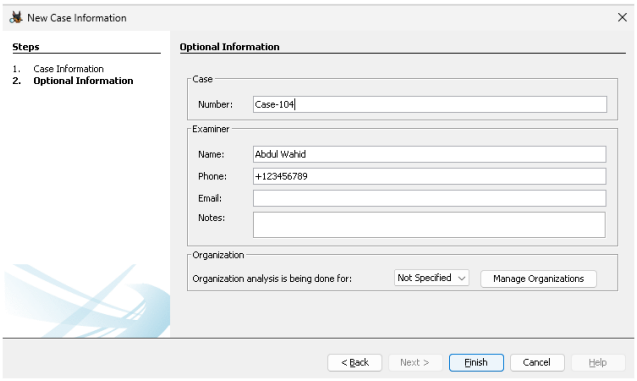
Case name
In the New Case Information window, set:

Case Name: File Extension Mismatch

Base directory
Set **Base Directory** to the Image File Analysis folder on the Desktop (C:\Users\Administrator\Desktop\Image File Analysis\).
Click **Next**.



Enter Case Number **104** and your examiner details → click **Finish**.



Select Host

Leave the default selection → click **Next**.

Data source type

Ensure **Disk Image or VM File** is selected → click **Next**.

Browse for the image

Click **Browse**.

Select the evidence file

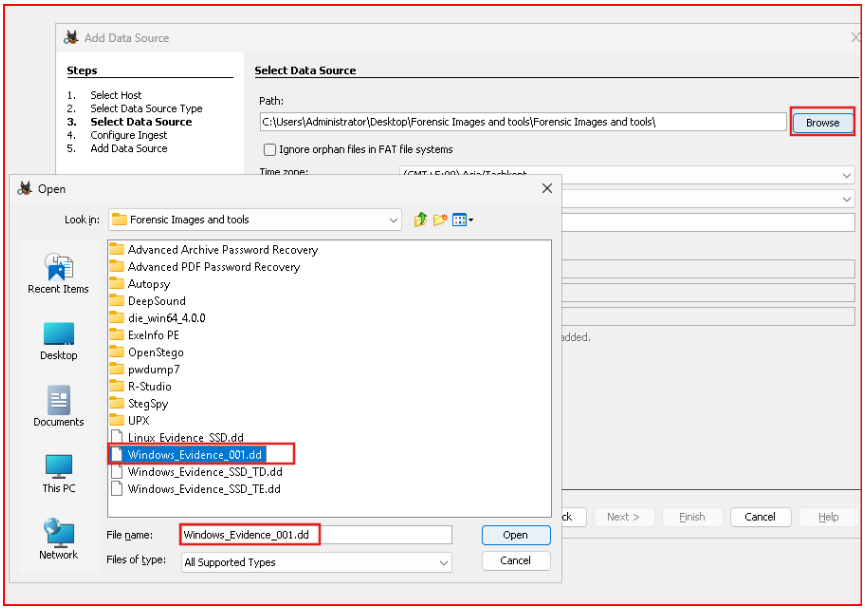
Navigate to:

E:\CHFI-Tools\Evidence Files\Forensic Images

Select **Windows_Evidence_001.dd** → click **Open**.

Confirm the path

The path appears in the Path field → click **Next**.



Configure ingest modules

In the **Run ingest modules on:** section, check (at minimum):

- File Type Identification
- Extension Mismatch Detector
- Embedded File Extractor

  Click **Next**.

Add the data source

When "Data source has been added to the local database. Files are being analyzed" appears, click **Finish**.

Let the scan run to completion (do not use the machine until it finishes).

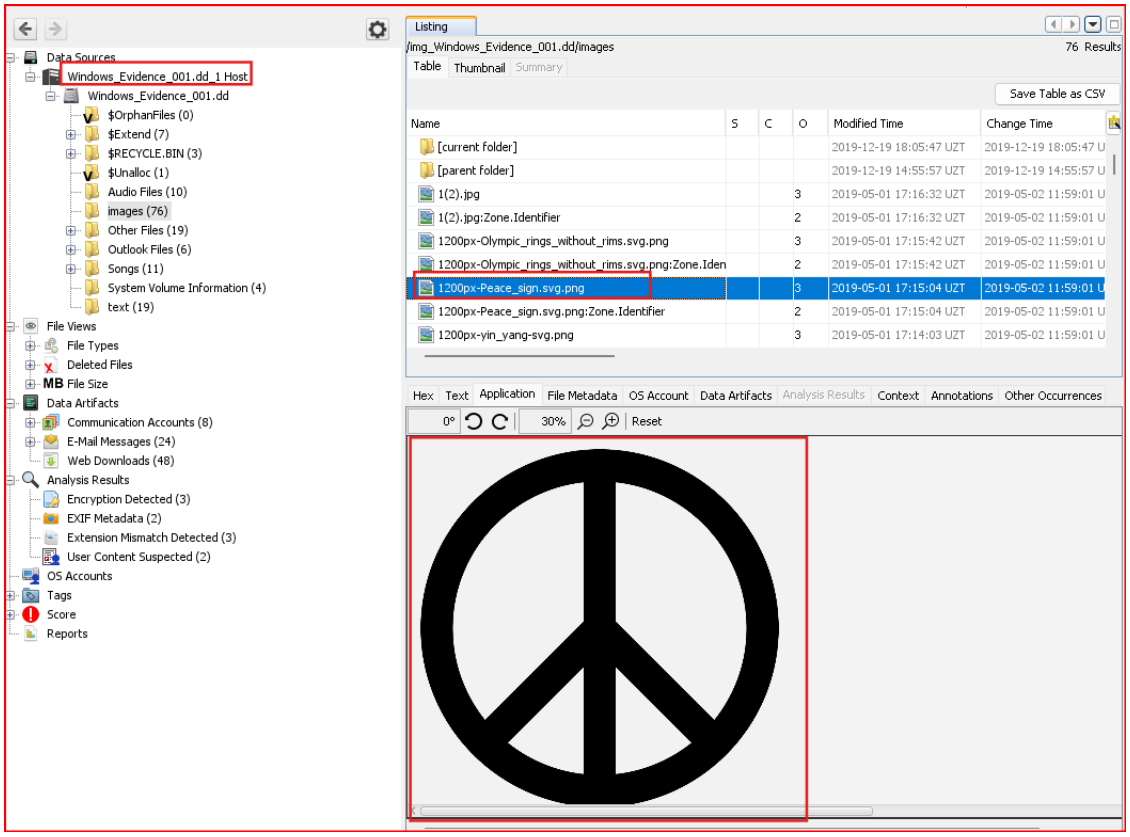
Expand Data Sources

In the left pane, expand **Data Sources**.

You will see **Windows_Evidence_001.dd** listed.

Open the image

Click the image filename to view its contents (normal OS folders) in the right pane.



View extension mismatches

In the left pane, under **Analysis Results**, click **Extension Mismatch Detected**.

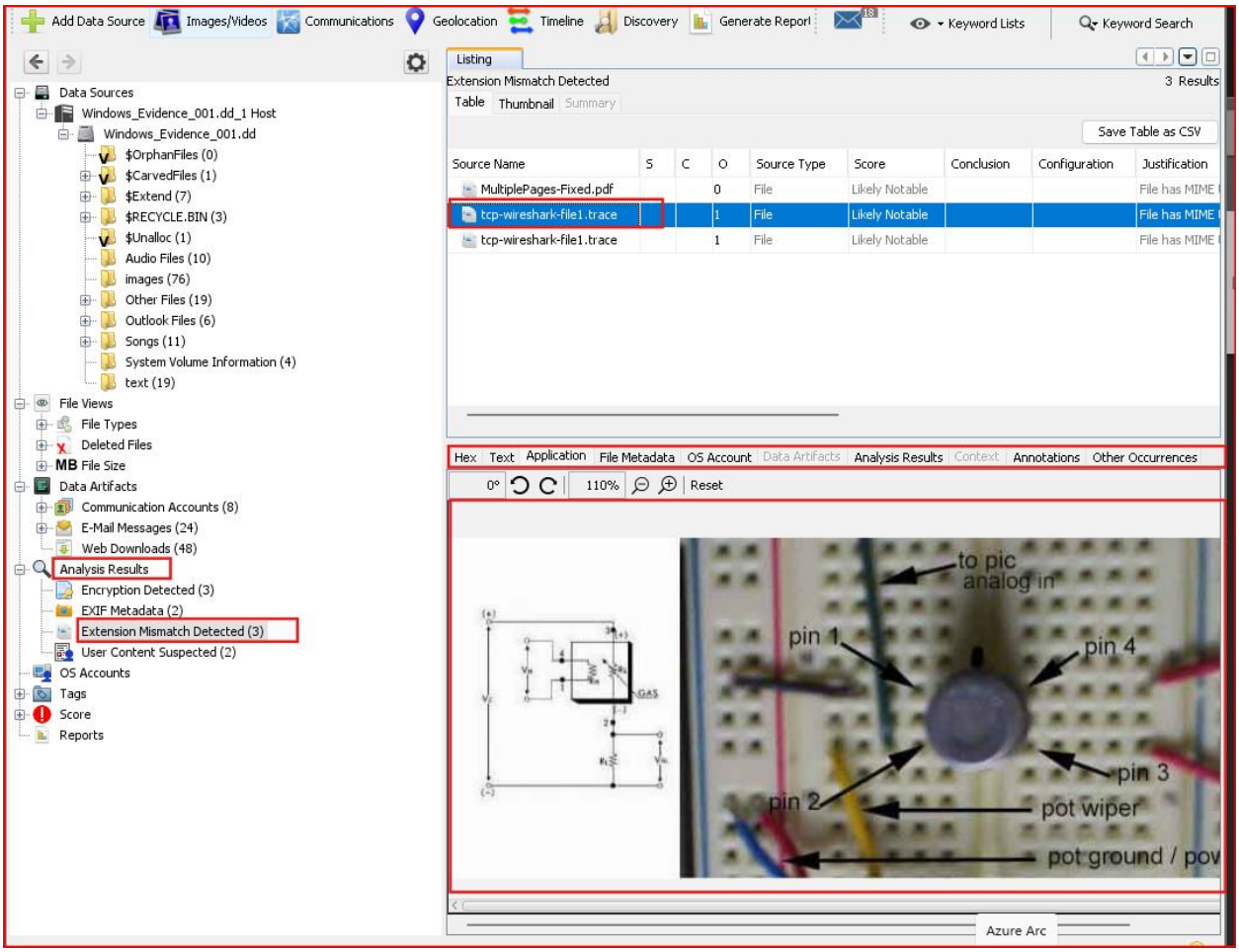
The right pane lists files whose content doesn't match their extension (usually 3 results).

Inspect a mismatched file

Select a file (example: tcp-wireshark-file1.trace).

Its content is shown in the lower pane — in the example it is actually a JPEG image of an electronic circuit, even though the extension is .trace.

You can also check the Hex, Text, File Metadata, and other tabs.



Close all open windows……

Unpack Program Packers

Lab Purpose:

The purpose of this lab is to demonstrate the methodology for identifying and defeating executable obfuscation techniques used as an anti-forensic evasion strategy. The lab provides hands-on experience in using static analysis tools like **Detect It Easy (DiE)** and **Exeinfo PE** to detect signature patterns of binary packers.

Lab Setup

Virtual machines needed

- Windows Server 2025 VM only (single machine)

Other requirements

- Administrator privileges
- CHFI-Tools available locally (confirm the drive letter — usually E:\CHFI-Tools)

Pre-checks

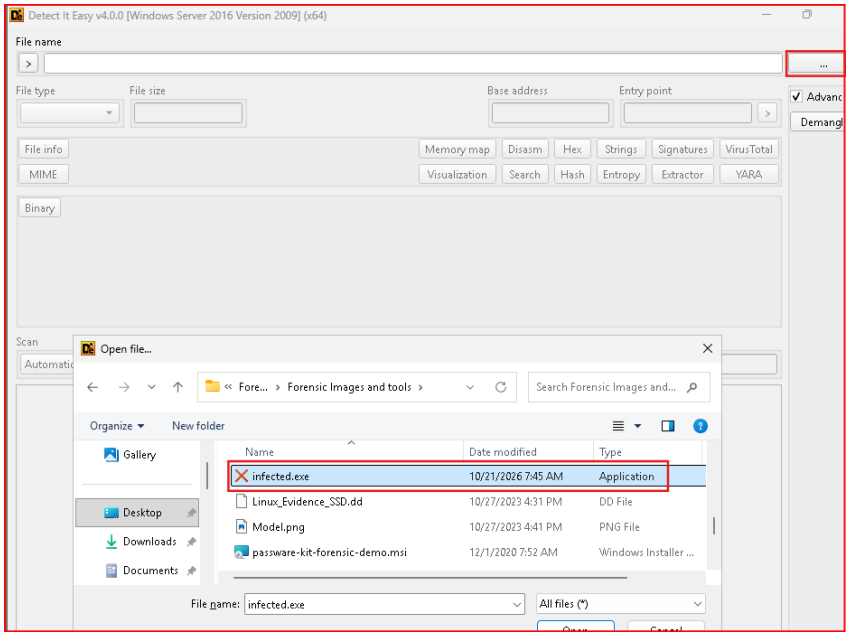
- You are logged in as Administrator.

Lab concept: A program packer compresses/obfuscates an executable so it can evade firewalls and antivirus tools. You first identify the packer used on **Infected.exe** with **Detect It Easy (DiE)** and **Exeinfo PE**, then unpack it with **UPX**.

Note: In this lab the malicious file is obviously named Infected.exe for demonstration. In a real case, attackers would give it a normal-looking name.

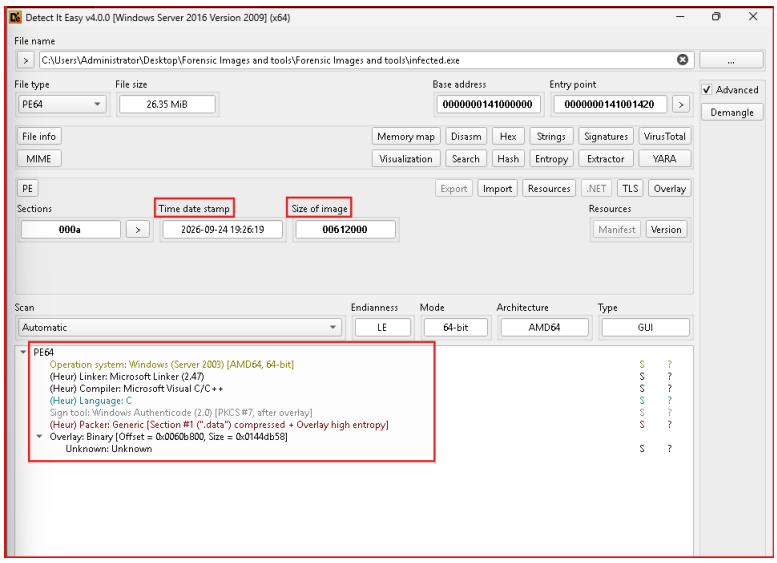
Identify the packer with Detect It Easy (DiE)

Select **infected.exe** → click **Open**.



Review the results

The tool displays:



Close Detect It Easy.

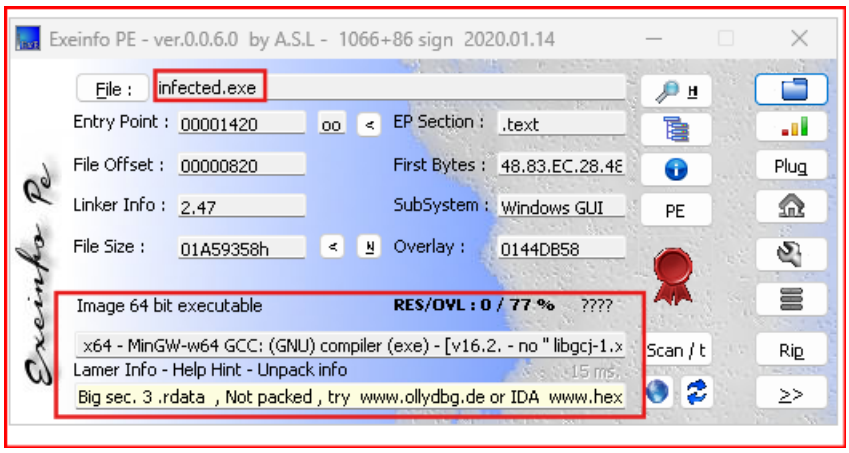
Exeinfo PE

Launch the tool
exeinfope.exe.

Open the target file
Click the **Open file** icon in the right pane.

Select the file

Review the results
The tool confirms:



Close Exeinfo PE.

Conclusion

This module shifted from passive file-system analysis to actively defeating anti-forensic countermeasures. TRIM proved to be a genuine anti-carving control — carving succeeded only when TRIM was off, on both Windows and Linux. Partition hiding by delete-and-merge was defeated through a raw sector scan rather than trusting the live partition table. Password protection on PDFs and archives was defeated through targeted brute-force/dictionary attacks once the likely character set was narrowed. Steganographic concealment in images and audio was defeated with purpose-built detection and extraction tools across both Windows and Linux. Extension masquerading and executable packing were both defeated the same way — by trusting binary signatures and entropy over file names, icons, and surface appearance. The common thread across all eight labs is that anti-forensic techniques manipulate metadata, structure, or surface presentation, while defeating them requires inspecting the underlying bytes directly. The main gap to close in later work is independent hash verification of every recovered/cracked artifact, which this module's procedure did not uniformly capture.

